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LEARNER GUIDE

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20.0	1 March 2025	Legacy master learner guide — Innovative Problem Solving with Generative AI.	Dr. Alfred Ang
21.0	17 August 2026	Major revision. Retitled to the published course title. Rebuilt around 10 real-life workplace case-study activities, each with scenario, evidence table, detailed step-by-step, GenAI prompt, discussion questions, debrief and self-check. Added expanded theory chapters for all three topics, a reusable 10-prompt library, assessment preparation and a 31-term glossary. Content beefed up from Skills for Success, IMD, MIT CCMIT, Coursera, Six Sigma, LSE Business Review and current generative-AI problem-solving research.	Dr. Alfred Ang

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Introduction

This Learner Guide accompanies the WSQ course WSQ - Generative AI for Problem Solving (TGS-2023036653), conducted by Tertiary Infotech Academy Pte Ltd. It is your step-by-step companion for all ten hands-on case-study activities, and your reference after the course when you take these methods back to your own workplace.

The course is built around a single idea: most organisations are very good at generating solutions and very poor at defining problems. The result is expensive activity that changes nothing. Over two days you will learn a complete, repeatable method — frame the problem, diagnose the true root cause, generate a wide solution set, converge on a defensible choice, implement it against real human resistance, and prove whether it worked.

Generative AI is used throughout, but never as an oracle. It is a divergence engine and a tireless challenger: it expands what you consider and accelerates how fast you get there. The evidence, the judgement and the accountability remain yours. Every activity therefore asks you to compare what the AI produced against what your evidence actually supports.

This course aligns to the SkillsFuture Singapore Skills Framework Technical Skill and Competency "Problem Identification" (RET-ACE-4006-1.1).

Course Learning Outcomes

- LO1: Identify performance deficiencies and root causes impacting organisational aspects
- LO2: Deduce key implications, develop corrective plans, and shortlist viable solutions
- LO3: Determine preferred solutions and evaluate the effectiveness of implemented plans

TSC Abilities Covered

- A1: Identify the types of performance deficiency and examine the causes and their impact on organisation-related aspects
- A2: Identify the root causes of the problems with team members using appropriate group facilitation techniques
- A3: Deduce relevant linkages and patterns to identify key implications on organisational systems and processes
- A4: Develop corrective action plans for any shortfalls identified in implemented solutions
- A5: Shortlist and evaluate the most viable ideas using appropriate problem-solving and decision-making techniques and tools
- A6: Determine a preferred solution using appropriate methods and draw up implementation plans
- A7: Evaluate the effectiveness of the implemented solution and implementation plans

TSC Knowledge Covered

- K1: Criteria for identifying performance deficiency or cause of failure in organisational systems and processes
- K2: Types of analytical tools and techniques in terms of problem identification
- K3: Application of problem solving tools and techniques
- K4: Techniques used during problem solving and decision making processes
- K5: Group facilitation techniques for root cause identification
- K6: Types of decision making models for arriving at the preferred solution and their features
- K7: Techniques to evaluate the effectiveness of implemented solution and implementation plan
- K8: Factors affecting the effectiveness of an implementation plan
- K9: Rationale for the different components in a corrective action plan

How to Use This Guide

- Each activity section is self-contained: the scenario, the evidence, the exact steps, the GenAI prompt, the discussion questions and the debrief.
- Work through the steps in order during the class. The steps are deliberately detailed so you can repeat the method at work without the trainer present.
- The prompts are written to be copied verbatim into ChatGPT, Microsoft Copilot, Google Gemini or Claude. Replace anything in <<double angle brackets>> with your own situation.
- The Self-check at the end of each activity tells you when your output is genuinely finished, rather than merely complete-looking.
- The assessment is open book: this guide and the slides are permitted. Internet search and AI tools are NOT permitted during assessment.

Tools You Will Use

Tool	URL	What it is for
5 Whys	https://alfredang.github.io/5whys/	Iterative why-laddering to drive from symptom to root cause.
Fishbone	https://alfredang.github.io/fishbone/	Ishikawa cause categorisation across People, Process, Technology, Material, Environment, Measurement.
Pareto Chart	https://alfredang.github.io/paretochart/	80/20 ranking of defect/complaint categories to target the vital few.
System Loop	https://alfredang.github.io/systemloop/	Causal loop mapping of reinforcing (R) and balancing (B) feedback loops.
CollabNote	https://alfredang.github.io/collabnote/	Shared note wall for posting team problem statements.
Pinboard	https://alfredang.github.io/pinboard/	Digital post-it wall for solution brainstorming and reflection.
Generative AI	ChatGPT / Copilot / Gemini / Claude	Framing, cause hypotheses, ideation, scoring and structuring evaluation.

Topic 1 — Identifying Performance Gaps and Root Causes

1.1 Why Problem Definition Comes First

Charles Kettering's line — "a problem well stated is a problem half solved" — is the most under-applied principle in management. A vague problem statement does not merely slow you down; it actively misdirects spending, because every department resolves the ambiguity in favour of the work it already wanted to do.

In the ShopFront SG case you will meet in Activity 1, one sentence — "customers are unhappy and sales are dropping" — sends Marketing to a discount campaign, IT to an app redesign and Customer Service to hiring temps. Ninety days and S$180,000 later, nothing has changed. Not one of those three teams did anything unreasonable. The failure was upstream of all of them.

Symptom, problem and root cause

	Symptom	Problem	Root cause
What it is	What people notice and complain about	The measurable performance deficiency	The system condition that produces it
ShopFront example	"Customers are unhappy"	Checkout completion fell from 68% to 51%	No capacity planning informed by analytics
Effect of fixing it	Feels responsive; changes nothing	Moves the number this quarter	Stops the problem recurring
Typical response	A campaign or a memo	A project	A change to policy, process or incentive

Treating a symptom always feels faster, which is precisely why it is the most expensive habit in problem solving. The discipline this course teaches is to spend deliberate, uncomfortable time at the problem and root-cause layers before authorising any spend.

Well-defined and ill-defined problems

A well-defined problem has a clear goal and a known method — "reduce app load time to under two seconds". An ill-defined problem has neither — "improve the customer experience". Most problems arrive at your desk ill-defined, which means framing is not preparation for the work; framing IS the work. Generative AI is particularly valuable here because it can rapidly restate an ill-defined problem in several different structured forms, letting you choose the framing that best fits your evidence.

The IDEAL cycle

The IDEAL heuristic (Bransford & Stein, taught at MIT) gives you a five-stage backbone that this entire course follows:

1. Identify the problem — recognise that a problem exists and pinpoint what it actually is, by asking questions and gathering information rather than assuming.
2. Define the context — investigate the facts. When did this last work correctly, and what changed?
3. Explore possible strategies — brainstorm and develop multiple solution pathways before committing.
4. Act on the best solution — implement the selected approach.
5. Look back and learn — evaluate the outcome and document the lesson.

Note: Most teams start at step 4. The cost of skipping steps 1 and 2 is always paid later, with interest, because you fix the wrong thing and then have to fix it again.

The six sub-skills of problem solving

The Government of Canada's Skills for Success framework defines problem solving as the ability to identify, analyse, propose solutions and make decisions. It breaks into six sub-skills, which map directly onto this course:

Sub-skill	What it means	Where you practise it
Issue identification	Is the problem familiar or novel, simple or complex?	Activity 1
Information gathering	Collect facts; find how similar problems were solved before	Activities 1-4
Analysis	Break the problem into parts; establish cause and effect	Activities 2-5
Action generation	Develop several solutions across short and long horizons	Activities 6-7
Implementation	Select and execute, staying flexible	Activities 8-9
Evaluation	Assess effectiveness and extract the lesson	Activity 10

1.2 Writing a Measurable Problem Statement

A workplace problem statement is not a description of a feeling. It is a contract that lets a team start work, lets a sponsor approve spend, and lets everyone evaluate the outcome later. It carries six elements, and all six are compulsory.

Element	What it captures	Failure if missing
Context	Who, where, and which process is affected	Scope creep; wrong people involved
Metric	The single primary measure you will move	Teams optimise different things
Baseline	Where the metric is today, with the number	You can never prove improvement
Target	Where it must reach, with the number	No definition of success
Timeframe	By when	Nothing is ever late
Constraints	Budget, headcount, compliance, brand	Solutions proposed that cannot be adopted

The Baseline element deserves particular attention. It is the element teams most often skip, and it is the one that makes evaluation possible sixteen hours later in Topic 3. If you take one habit from this course, take this: never accept a problem statement without a baseline number.

Worked example

Weak: "Our customers are not happy with our services."

Strong: "Checkout completion on the ShopFront SG app and web store has fallen from 68% to 51% (benchmark 70%) over 12 months, alongside app load times rising from 2.1s to 4.8s, contributing to a S$0.9m per month decline in online sales. We aim to restore checkout completion to at least 68% and app load to under 2.0s within 90 days, without additional headcount and without changes requiring PDPA re-consent."

1.3 Cognitive Barriers

Problem solving fails for predictable psychological reasons as often as for analytical ones. Know these six and you will catch yourself in the act:

Barrier	How it shows up at work
Functional fixedness	Seeing a resource only for its usual purpose, so obvious re-uses stay invisible
Mental set	Re-using the solution that worked last time on a problem that has since changed
Confirmation bias	Collecting only the evidence that supports the cause you already suspect
Unnecessary constraints	Treating habits as rules; most 'fixed' constraints are never enforced by anyone
Irrelevant information	Drowning in data while the three numbers that matter go unexamined
Consensus fatigue	Accepting a weak option because the meeting has gone on too long

Note: Two of these — cognitive fixedness and consensus fatigue — are specifically named by IMD as the biases that most reduce the quality of leadership problem solving. Activity 6 is designed to break the first; Activity 7 is designed to defeat the second.

1.4 Generative AI in Problem Solving

Generative AI does not solve your problem. It expands what you consider and accelerates how fast you get there. Understanding exactly where it is strong and where it is weak is the difference between a productivity gain and a confidently-wrong decision.

Dimension	GenAI is strong	GenAI is weak
Breadth of options	Generates 20 ideas in seconds, free of office politics	Cannot tell which are real
Framing	Restates vague problems into structured, testable form	Invents specifics your evidence never supplied
Analogy	Borrows working patterns across every industry it has read	May transfer a pattern that does not apply

Arithmetic	Improving, and good at showing method	Still makes errors — verify every number
Context	Applies general best practice competently	Does not know your people, history or politics
Accountability	Tireless, patient, unbiased by hierarchy	Cannot be held responsible for the decision

Why GenAI hallucinates specifics — and what to do about it

A language model is trained to produce plausible continuations. When your prompt lacks a detail the answer seems to require, the model supplies a plausible one rather than leaving a gap. This is not a malfunction; it is the mechanism working as designed. In problem solving it is dangerous because invented specifics look identical to established facts.

The defence is procedural, not technical. Always supply your real evidence in the prompt, and always instruct the model to mark what it assumed. You will practise this in every single activity, and it is the most valuable AI habit you will take away from this course.

Known limitations from current research

- Poor mathematical reliability without external tools — always check the arithmetic yourself.
- Weak memory, planning and reasoning in single-agent use — break complex problems into steps.
- Unreliable on domain-specific questions without access to your proprietary data.
- Retrieval-augmented generation (RAG) and tool use materially improve reliability, which is why enterprise deployments connect models to real company data rather than relying on training data.

Prompting habits that work

- Give it a role — "Act as a root cause analysis expert" sets the reasoning frame.
- Supply the evidence — paste your real numbers; without them the model invents plausible ones.
- Demand structure — name the fields you want back, so outputs are comparable across teams.
- Ask it to flag assumptions — "mark anything you assumed that my evidence did not support".
- Make it challenge you — "ask me for evidence rather than accepting my answer".
- Iterate — the first output is a draft. The third is usually useful.

1.5 The Root Cause Toolkit

Four tools, four different jobs. Mature problem solvers use them together rather than choosing between them:

Tool	What it gives you	Use it when	Limitation
5 Whys	Depth on one causal chain	The problem is focused and singular	Follows only one branch
Fishbone	Breadth across all dimensions	Causes are many and unclear	Shallow on each branch
Pareto	Priority by contribution	You have counted, categorised data	Ranks frequency, not severity
System Loops	Feedback dynamics over time	Fixes keep rebounding	Needs time-series insight

Note: The tools chain naturally: Pareto tells you WHERE to look, 5 Whys tells you WHY it happens, Fishbone makes sure you have not missed a dimension, and System Loops explains why your last fix stopped working.

Topic 2 — Developing Corrective Plans and Evaluating Solutions

2.1 Divergence Before Convergence

Generating and judging are different cognitive modes, and running them at the same time destroys your best ideas. A team that evaluates while it generates typically stops at six to eight conventional options, because every unusual idea is killed by the first objection before it can be developed.

The solution funnel therefore runs in strict sequence: diverge widely, cluster by theme, screen quickly, score rigorously, then commit. Each stage has its own tool, and mixing them is the most common facilitation error in workplace problem solving.

2.2 Techniques for Generating Solutions

Technique	How it works	Best for
Brainstorming	All ideas welcome, judgement suspended, quantity first	General divergence
SCAMPER	Seven prompts applied to a fixed assumption	Breaking cognitive fixedness
Generic Parts Technique	Strip the problem to abstract functions	Finding solutions in other domains
Analogy / cross-industry	Borrow a working pattern from another sector	Novel problems
Reverse brainstorming	Ask how to make it worse, then invert	Stuck teams
Nominal group technique	Silent individual generation, then group discussion	Avoiding groupthink
Six Thinking Hats	Examine from facts, feelings, risks, benefits, creativity, process	Balanced review
Mind mapping	Radiate ideas from the central problem	Seeing relationships

SCAMPER in detail

Move	The question it asks

Substitute	What can be swapped out — different person, material, place or rule?
Combine	What can be merged — two steps, two roles, two services?
Adapt	What works elsewhere that we could borrow?
Modify	What if we magnified, shrank or reshaped it?
Put to another use	What else could this resource or step do?
Eliminate	What if we simply removed it? (Often the biggest unlock.)
Reverse	What if we ran it backwards, or swapped who does what?

Hard constraints versus habits

The single most valuable output of a SCAMPER session is discovering that a constraint everyone treated as fixed is actually a habit. In the Meridian Health case, the MOH phlebotomy competency standard is a genuine hard constraint — it is a regulation with an enforcing body and a penalty. The 07:00-09:30 draw window at the centre is a habit; it feels equally fixed, but it exists only because nobody has questioned it.

The test is simple and you should apply it every time: who enforces this, and what is the penalty for breaking it? If you cannot name an enforcer, you are looking at a habit.

2.3 Converging on a Decision

Convergence needs explicit criteria, or the decision defaults to whoever is most senior or most persistent. Use two tools in sequence — a fast screen, then a rigorous instrument.

The Impact-Ease matrix

Quadrant	Meaning	What to do
Quick Win	High impact, easy to implement	Do these first — they build momentum and credibility
Big Bet	High impact, hard to implement	Do these, but plan and resource them properly
Fill-In	Low impact, easy to implement	Do only if you have spare capacity
Thankless Task	Low impact, hard to implement	Do not start these

The weighted decision matrix

The Impact-Ease screen is directional. When you must defend a decision to a board, you need a weighted matrix: explicit criteria, explicit weights, scores from 1 to 5, and a computed total.

The critical facilitation move is to agree the WEIGHTS before scoring anything. Most arguments that appear to be about scores are really disagreements about weights, and surfacing that early converts a political argument into an explicit, recorded choice.

Note: Always run a sensitivity test. Change one weight materially and re-rank. If the winner changes, your recommendation is fragile and you must say so. If it holds across several weightings, you have a robust recommendation — and boards trust robust recommendations.

2.4 The Corrective Action Plan

A corrective action plan converts a chosen solution into something a colleague could execute without you in the room. Eight components, each of which exists because its absence causes a specific, predictable failure:

Component	Why it exists	Failure if missing
Root cause addressed	Ties the action to the diagnosis	Solution drifts back to treating a symptom
Corrective action	The specific thing being done	Vague intent that never becomes execution
Owner (named role)	Single point of accountability	Everyone assumes someone else is doing it
Timeline / milestone	Makes progress checkable	Slips indefinitely and nobody notices
Resources required	Budget, people and tools committed	Stalls at the first resource conflict
Success measure	Defines what 'worked' means	Cannot evaluate; endless debate
Risk and mitigation	Anticipates what breaks it	Surprised by an entirely predictable failure
Review checkpoint	Forces a continue/stop decision	Runs on long past the point of failure

Note: An owner must be a named role, never a department. 'Operations' cannot be phoned, cannot be held to a date and cannot be asked why something slipped. 'Centre Operations Manager' can.

Topic 3 — Selecting, Implementing and Measuring Effectiveness

3.1 Why Good Solutions Fail

Most solutions fail on adoption rather than on design. The organisation in Activity 9 bought a S$400,000 CRM that worked perfectly and that nobody used. The technology was not the problem. This is why the Skills Framework treats 'factors affecting the effectiveness of an implementation plan' (K8) as a distinct body of knowledge: implementation is mostly about people.

Factor	What good looks like
Sponsorship	A senior owner who can re-prioritise competing work, not merely endorse yours
Sequencing	Dependencies respected — never drive demand into a system you have not yet fixed
Capacity	Teams already at capacity are given real relief, not encouragement
Stakeholder buy-in	Resistance treated as a design input, surfaced early
Communication	Regular, specific, and continuing past launch

Measurement	Window, baseline and control defined at day zero

3.2 Reading and Working With Resistance

Resistance is information. It usually tells you something true about your plan that you did not know. Learning to read what someone is actually objecting to — as opposed to what they say — is the core skill of implementation.

Stakeholder type	What they actually object to	The move that works
At-capacity team	The workload, not the idea	Real relief — re-prioritise other work or fund capacity
Face at risk	Being shown to have been wrong	Reframe rather than relitigate; give them ownership of a win
External party	Cost and scrutiny	Co-design the standard, then tie it to commercial consequence
Job-security fear	A rational threat to their livelihood	Commit only to what you can actually guarantee
The indifferent	Nothing at all — which is the danger	Give them a stake before you need them

Note: The messenger matters as much as the message. The same words land completely differently coming from the sponsor, a peer, or the analyst who produced the diagnosis. Assign a messenger to every difficult conversation and be able to justify the choice.

Indifference kills more rollouts than opposition does, because nobody escalates an absence of enthusiasm. Opposition is visible and gets managed; indifference is invisible until launch day.

Change management actions

- Leadership alignment — communicate the business goal and tie measurable KPIs to outcomes.
- Quick wins first — deliver an early visible improvement to build momentum and credibility.
- Involve affected teams early, in solution design rather than after the decision.
- Co-design with likely resisters — an imposed standard is complied with on paper; a co-designed one is actually adopted.
- Incentives and accountability — link the change to metrics people are genuinely measured on.
- Train and support — SOPs, quick guides and a real support channel; adoption dies without them.

3.3 Measuring Effectiveness

"It's working" is not an evaluation. A credible evaluation gives a verdict per metric against both baseline and target, states attribution explicitly, and is honest about what cannot be determined from the available evidence.

Leading and lagging indicators

	Leading indicators	Lagging indicators
Respond in	Days to weeks	Months to quarters
Examples	Page load time, checkout completion, adoption rate	CSAT, repeat purchase, brand trust
They tell you	Whether the change is taking hold	Whether the outcome actually improved
Risk if used alone	An early signal that may not persist	Feedback arrives too late to steer
How to use them	Steer with these week to week	Set realistic horizons and judge outcomes

Expecting a lagging indicator to move inside 90 days is a planning error, not an execution failure. Customer satisfaction reflects accumulated experience across several purchase cycles — for a fashion retailer, six to nine months rather than three.

Attribution and confounding

In a real workplace you rarely get a clean experiment. Something else always happens during your measurement window — a seasonal peak, a competitor's move, another internal project. Good practice is to define the measurement window at day zero, hold a control where possible (for example a staged rollout by region), and segment the data afterwards.

Where you cannot establish attribution, say so plainly. Stating what you cannot determine is what makes the rest of your evaluation credible to a finance director.

Solution failure versus implementation failure

	Solution failure	Implementation failure
What happened	It was genuinely done, and it did not work	It was never really done
Evidence to check	Open rates, click rates, sector benchmarks	Adoption and compliance rates
Right response	Change or drop the solution	Fix the rollout and the change management
Common error	Abandoning something that was never adopted	Blaming people for a bad design

Note: This distinction decides where the next tranche of budget goes. Abandoning a solution that was never actually adopted throws away a fix that might have worked perfectly well.

Sustaining the gain

- PDCA / DMAIC-Control — without a control phase, improvements decay back to the old behaviour.
- Standardise — write the new way into the SOP, the induction and the checklist.
- Monitor — keep the leading indicator on a dashboard someone actually looks at.

- Assign ownership — a named role owns the sustained metric after the project team disbands.
- Review cadence — a standing checkpoint that carries a decision, not a status update.
- Capture the lesson — the 'Look back and learn' step of IDEAL, and the one most often skipped.

Hands-On Activities

Each activity below is a real-life workplace case study. Work in the team size stated, follow the steps in order, run the prompt, discuss the questions, and use the debrief to check your thinking. The Self-check tells you when your output is genuinely finished.

Activity 1 — Rewriting a Vague Problem into a Measurable Problem Statement

Objective

A1 · K1 — Identify the type of performance deficiency and express it as a measurable problem statement.

At a glance

Field	Detail
Case study	ShopFront SG — 'Customers are unhappy and sales are dropping'
Duration	45 minutes
Grouping	Teams of 3
Tools	ChatGPT / Copilot / Gemini, CollabNote
Ed-tool	CollabNote — https://alfredang.github.io/collabnote/

The scenario

ShopFront SG is a 42-outlet fashion retailer with an e-commerce site and a mobile app. At the Monday management meeting the Head of Retail Operations opens with one line: "Customers are unhappy and sales are dropping — fix it." No metric, no baseline, no scope. Three departments leave the room with three different interpretations. Marketing books a S$180,000 discount campaign. IT starts an app redesign. Customer Service hires two temps. Ninety days later the complaint rate is unchanged and the budget is gone.

You are the business analyst asked to restate the problem BEFORE any more money is committed. The dashboard extract below is all the evidence you have.

The evidence

ShopFront SG — 90-day performance extract

Metric	12 months ago	Current	Industry benchmark
Average app page load	2.1 s	4.8 s	under 2.0 s
Checkout completion rate	68%	51%	70%
Cart abandonment	61%	74%	60%
CSAT (post-purchase)	84%	70%	85%
Repeat purchase (90-day)	38%	29%	40%
Complaint tickets / week	120	410	-
In-store sales	S$4.2m/mo	S$4.1m/mo	-
Online sales	S$2.8m/mo	S$1.9m/mo	-

Step-by-step

6. Read the ShopFront SG scenario and the 90-day evidence table as a team.
7. Individually and in silence, underline the one metric you believe is the real performance deficiency. Do not discuss yet — this prevents anchoring on the loudest voice.
8. Compare choices around the table. Where you disagree, argue from the numbers only.
9. Draft the problem statement on paper with all six elements labelled: Context, Metric, Baseline, Target, Timeframe, Constraints.
10. Open your GenAI tool and run the Problem Statement prompt supplied on the slide.
11. Compare the AI output against your draft. Highlight every specific the AI supplied that the evidence did not support — label each one 'ASSUMPTION'.
12. Merge the two into a final statement your team can defend to the Head of Retail Operations.
13. Post the final statement to the shared wall at https://alfredang.github.io/collabnote/
14. Nominate one speaker to read it out in 60 seconds and name the one assumption you are least comfortable with.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
You are a business analyst for a Singapore fashion retailer.

Refine the vague statement below into ONE clear, measurable problem statement.

Vague statement: "Customers are unhappy and sales are dropping."

Evidence:
- App page load rose from 2.1s to 4.8s (benchmark <2.0s)
- Checkout completion fell 68% -> 51% (benchmark 70%)
- Cart abandonment rose 61% -> 74%
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- CSAT fell 84% -> 70%
- Online sales fell S$2.8m -> S$1.9m per month; in-store flat

Your problem statement MUST contain, each labelled:
- Context (who, where, which process)
- Metric (the single primary measure)
- Baseline (where it is now, with the number)
- Target (where it must be, with the number)
- Timeframe (by when)
- Constraints (budget, headcount, compliance, brand)

Then list any assumption you had to make, and state what additional evidence
would let you remove that assumption. Do not propose solutions yet.
```

Discussion questions

15. Which metric in the table is the SYMPTOM the executive noticed, and which metric is closest to the actual performance deficiency? Justify with the numbers.
16. The evidence shows in-store sales are flat while online sales fell 32%. What does that single contrast rule out as a cause, and why does that matter before you spend a dollar?
17. Write the problem statement with all six elements. Which element was hardest to fill, and what does the gap tell you about your organisation's data?
18. Compare your team's statement with the GenAI-generated one. What did the AI assume that is NOT supported by the evidence? (Every team will find at least one.)
19. Whose sign-off do you need on this problem statement before work starts, and what would you show them to get it?

Debrief

EXPECTED DIRECTION — the deficiency is a DIGITAL CHANNEL conversion failure, not a general 'unhappy customer' problem. In-store being flat while online fell 32% localises the problem to the digital funnel and rules out brand, product and pricing as the primary cause — which immediately invalidates Marketing's S$180k discount campaign.

A defensible statement reads roughly: 'Checkout completion on the ShopFront SG app and web store has fallen from 68% to 51% (benchmark 70%) over 12 months, alongside app load times rising from 2.1s to 4.8s, contributing to a S$0.9m per month decline in online sales. We aim to restore checkout completion to at least 68% and app load to under 2.0s within 90 days, without additional headcount and without changes requiring PDPA re-consent.'

KEY TEACHING POINTS:
1. The executive named a symptom ('unhappy'). CSAT is a lagging perception measure; checkout completion is the operational deficiency you can actually act on.
2. Six elements are non-negotiable. A statement missing a Baseline cannot be evaluated later — this is exactly what Topic 3 comes back to.
3. GenAI reliably invents plausible specifics (a peak-hour window, a payment gateway name, a customer segment) that the evidence never supplied. Learners must mark

these as assumptions. This is the single most important AI-literacy habit in the course.
4. Note what has NOT been established yet: WHY load time rose. That is root cause, and it is Activity 2. Resist the team that jumps to 'we need more servers'.

Self-check — is your output finished?

Your statement is complete when a colleague who was NOT in the room can read it and correctly state (a) which metric is being fixed, (b) the number it is at today, (c) the number it must reach, (d) by when, and (e) what you are not allowed to change — without asking you a question. If any of the five is missing, the statement is not finished.

Activity 2 — 5 Whys Root Cause Analysis

Objective

A2 · K2, K5 — Identify root causes with team members using structured group facilitation.

At a glance

Field	Detail
Case study	Meridian Health Screening — the 07:00 blood-test backlog
Duration	45 minutes
Grouping	Teams of 3-5
Tools	5 Whys ed-tool, ChatGPT / Copilot / Gemini
Ed-tool	5 Whys Tool — https://alfredang.github.io/5whys/

The scenario

Meridian Health operates six health-screening centres in Singapore. Corporate clients book annual screening packages; the fasting blood draw must happen between 07:00 and 09:30.

Over the last four months, 31% of morning appointments overrun by more than 40 minutes. Corporate clients have escalated twice. One major client (1,400 employees, S$310,000 annual contract) has given notice of review. The Operations Manager's proposal is to open a seventh centre at a capital cost of S$1.2m.

The Clinical Director is unconvinced. She has asked your team to establish the ROOT CAUSE before the board approves any capital expenditure. The nurses tell you the same thing every morning: "We're just short-staffed." The data below says something more interesting.

The evidence

Meridian Health — morning screening operations, 4-month sample

Observation	Value
Booked slots 07:00-09:30	48 per centre
Phlebotomists rostered 07:00	3 of 5
Phlebotomists rostered 08:30	5 of 5
Mean draw time, trained staff	6 min
Mean draw time, relief staff	14 min
Relief staff share of morning roster	40%
Repeat draws (failed first attempt)	18%
Clients arriving without fasting compliance	12%
Lab courier pickup (fixed)	09:45
Samples missing the 09:45 courier	23%
Permanent staff turnover, 12 months	44%

Step-by-step

20. Read the Meridian Health scenario and study the operations table.
21. Appoint a facilitator (asks the whys and keeps the team on one branch) and a scribe.
22. Open the 5 Whys ed-tool at https://alfredang.github.io/5whys/
23. Enter the problem statement in the tool's problem field, using numbers from the table.
24. Ask Why #1. The facilitator's job: reject any answer that is an opinion, and ask 'what evidence supports that?'
25. Continue to five levels, entering each why and answer in the tool. Stay on ONE causal branch.
26. Mark the level where the answer stopped being a description of events and became a system explanation.
27. Run the same problem through the GenAI 5 Whys prompt from the slide, answering as the clinical team would.
28. Compare the two chains. Where they diverge, decide which branch the EVIDENCE supports and record why.
29. Write your root cause statement in one sentence, and note which levels remain unverified.
30. Present to the room in 90 seconds: the chain, the root cause, and your verdict on the S$1.2m proposal.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as a root cause analysis expert facilitating a clinical operations team.

Problem: 31% of morning health-screening appointments at Meridian Health
overrun by more than 40 minutes, and 23% of blood samples miss the fixed 09:45
lab courier.

Evidence: only 3 of 5 phlebotomists are rostered at 07:00 (all 5 by 08:30);
relief staff take 14 min per draw vs 6 min for trained staff; relief staff are
```

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40% of the morning roster; 18% of draws need a repeat attempt; permanent staff
turnover is 44% a year.

Apply the 5 Whys technique. Ask ONE why at a time and wait for my answer before
asking the next. Go five levels deep.

At each level, state what EVIDENCE would confirm or refute that level — do not
accept my answer at face value. If my answer is an opinion rather than a fact,
say so and ask me for the data that would support it.

After level 5, summarise: the root cause, the evidence supporting it, and which
levels remain unverified assumptions.
```

Discussion questions

31. Run the 5 Whys to five levels. At which level did you stop describing WHAT happens and start explaining WHY the system produces it?
32. The nurses say 'we're short-staffed'. The data shows all 5 phlebotomists are rostered by 08:30. What is the real constraint — headcount, or something else?
33. Trace the causal chain to 44% annual turnover. If turnover is the root cause, what does that say about the S$1.2m seventh centre proposal?
34. Where could this 5 Whys chain have gone wrong? Identify one point where a different, equally plausible 'why' would have led you somewhere completely different.
35. The GenAI facilitator challenged at least one of your answers as opinion rather than fact. Which one, and were you able to supply the evidence?

Debrief

EXPECTED CHAIN (teams may reach this by slightly different routes — accept any chain that is evidence-supported and reaches a systemic cause):
Why 1: Appointments overrun -> draws take longer than the slot allows.
Why 2: Draws take longer -> 40% of morning staff are relief staff at 14 min vs 6 min, and 18% of draws need a repeat.
Why 3: So many relief staff -> only 3 of 5 permanent phlebotomists are rostered at 07:00.
Why 4: Under-rostered at 07:00 -> permanent staff turnover is 44%/yr; vacancies are backfilled with relief staff who are not trained to the centre's protocol.
Why 5: Turnover is 44% -> the early shift is unattractive and there is no structured onboarding or retention path for phlebotomists.

ROOT CAUSE: a workforce retention and onboarding failure, surfacing as a capacity problem.

KEY TEACHING POINTS:
1. The S$1.2m seventh centre solves the SYMPTOM. A new centre staffed by the same 40% untrained relief pool reproduces the same overrun — you buy the problem twice.
2. 'We're short-staffed' was the team's stated cause and it was WRONG in the way that matters: they are not short of headcount, they are short of TRAINED headcount at

07:00. This distinction is worth S$1.2m.
3. Facilitation discipline: one why at a time, and every level must survive 'what evidence supports this?'. The GenAI acting as challenger models the facilitator behaviour learners should copy — this is the K5 group facilitation outcome.
4. Note the branch point: at Why 2 a team could have followed the 12% fasting non-compliance instead and landed on 'client communication'. That is a legitimate SECOND cause. 5 Whys gives depth on one branch — which is exactly why Activity 3 uses Fishbone for breadth.

Self-check — is your output finished?

Your root cause is sound when you can answer YES to all three: (1) If this cause were removed, would the 31% overrun stop recurring — not just improve this month? (2) Is every level in the chain supported by a number in the evidence table, or explicitly flagged as an assumption? (3) Does the root cause point at a SYSTEM (policy, process, incentive) rather than at a person? If your chain ends at 'the relief staff are slow', you have found a symptom and blamed a human — go two levels deeper.

Activity 3 — Fishbone (Ishikawa) Diagram for Breadth of Causes

Objective

A2 · K2, K5 — Categorise potential causes across all contributing dimensions.

At a glance

Field	Detail
Case study	Horizon Bank — branch service complaints at Jurong East
Duration	45 minutes
Grouping	Teams of 3-5
Tools	Fishbone ed-tool, ChatGPT / Copilot / Gemini
Ed-tool	Fishbone Tool — https://alfredang.github.io/fishbone/

The scenario

Horizon Bank's Jurong East branch has moved from best to worst in the retail network on customer experience in eleven months. Complaints run at 60 per month against a network average of 14. Average counter wait time is 58 minutes at peak; the service standard is 15.

Head Office ran a 'Service Excellence' refresher for all branch staff four months ago. Complaints did not move. The Branch Manager is now under performance review and morale is visibly poor — two tellers have resigned this quarter.

The Regional Director suspects the training was aimed at the wrong cause. Your team has been asked to map ALL contributing causes across every dimension before another intervention is funded. 5 Whys gave depth on one branch; this problem is too broad for one branch.

The evidence

Horizon Bank Jurong East — observation log and system data

Dimension	Observation
Queue	Single queue for all transaction types; no triage between simple and complex
Peak	11:30-14:00 accounts for 61% of daily footfall (lunch crowd from nearby offices)
Counters	6 counters exist; mean 3.2 staffed at peak
Systems	New KYC platform launched 11 months ago; teller must key client data into 3 systems
Systems	Mean core-banking screen response 8s, was 2s before the KYC platform
Staff	44% of tellers have under 6 months tenure
Staff	Teller KPI is transactions-per-hour; no customer-experience measure
Customer	34% of customers arrive without required documents
Customer	Branch signage and the app's document checklist do not match
Process	Complex cases (loans, disputes) occupy a counter for a mean of 27 minutes
Management	No structured feedback route from branch to Head Office product owners
Environment	Only 18 seats in the waiting area; standing customers visibly agitated

Step-by-step

36. Read the Horizon Bank scenario and the observation log.
37. Open the Fishbone ed-tool at https://alfredang.github.io/fishbone/
38. Enter the problem statement as the fish head, with the 60-complaints and 58-minute numbers.
39. Label the six bones: People, Process, Technology, Material/Information, Environment, Measurement.
40. Work bone by bone as a team. For each observation in the log, decide which bone it belongs on and phrase it as a CAUSE, not a complaint.
41. Mark every cause [EVIDENCED] if a number in the log supports it, or [HYPOTHESIS] if it is your team's inference.
42. Run the GenAI Fishbone prompt from the slide and add any bone entries your team missed.
43. Draw an arrow between any two causes on DIFFERENT bones where one drives the other. Note how many arrows you find.

44. Count the evidenced causes per bone. Identify which bone Head Office's training actually targeted.
45. Circle the three causes you would investigate first and write one line of justification for each.
46. Screenshot or export the diagram and present your top three plus your verdict on the training decision.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as a quality improvement facilitator running an Ishikawa (fishbone)
analysis with a bank branch team.

Problem (the fish head): Customer complaints at the Horizon Bank Jurong East
branch have risen to 60 per month (network average 14), with peak counter wait
time at 58 minutes against a 15-minute standard.

Generate candidate causes under these six bones: People, Process, Technology,
Material/Information, Environment, Measurement.

Rules:
- Give 4-6 candidate causes per bone, each phrased as a CAUSE not a complaint
- Mark each cause [EVIDENCED] or [HYPOTHESIS] based only on what I give you
below
- Do NOT propose solutions
- Finish by naming the 3 causes you would investigate first and say why

Evidence available: single queue with no triage; 61% of footfall between 11:30-
14:00; 3.2 of 6 counters staffed at peak; new KYC platform requires triple data
entry; screen response degraded 2s to 8s; 44% of tellers under 6 months tenure;
teller KPI is transactions-per-hour only; 34% of customers arrive without
documents; app checklist and branch signage disagree; complex cases take 27
minutes at the counter; no branch-to-HO feedback route; 18 seats in the waiting
area.
```

Discussion questions

47. Populate all six bones. Which bone filled up fastest, and which was hardest? What does an empty bone usually mean — no causes, or no visibility?
48. Head Office's answer was a Service Excellence training refresher. Which bone does that intervention target, and what share of your evidenced causes sit on that bone?
49. Find a cause on your diagram that CREATES another cause on a different bone. (Hint: look at the KPI and the queue.) What does that tell you about treating bones as independent?
50. Which three causes would you investigate first? Defend the ranking on evidence strength and likely contribution — not on ease of fixing.

51. The teller KPI is transactions-per-hour. Predict the behaviour that KPI produces at a counter when a confused elderly customer needs 20 minutes. Which bone does that belong on?

Debrief

EXPECTED DISTRIBUTION — a well-built diagram is heavily loaded on Process, Technology and Measurement, and only lightly on People:
PEOPLE — 44% of tellers under 6 months; two resignations; low morale; thin coaching.
PROCESS — single queue, no triage; complex cases (27 min) block counters; rostering does not match the 11:30-14:00 peak (3.2 of 6 counters staffed).
TECHNOLOGY — KYC platform forces triple data entry; screen response degraded 2s to 8s.
MATERIAL/INFORMATION — app checklist contradicts branch signage; 34% arrive without documents.
ENVIRONMENT — 18 seats for a 58-minute wait; standing customers escalate faster.
MEASUREMENT — transactions-per-hour KPI with no experience measure; no branch-to-HO feedback loop.

KEY TEACHING POINTS:
1. THE HEADLINE: Head Office trained the PEOPLE bone, which carries the fewest evidenced causes. That is precisely why complaints did not move and S$ was wasted. Learners should leave able to say: 'we intervened on the wrong bone.'
2. CROSS-BONE CAUSATION: the transactions-per-hour KPI (Measurement) drives tellers to rush or avoid complex cases (Process), which pushes complex customers back into the queue (Process), which lengthens the wait (Environment), which raises complaint volume. Bones are NOT independent — this observation is the bridge into System Loops in Activity 5.
3. FISHBONE vs 5 WHYS: 5 Whys gave one deep chain; Fishbone gives six shallow ones. Depth without breadth means you fix one cause and the problem persists; breadth without depth means you fix symptoms everywhere. Mature practice uses both — and the deliberate contrast between Activity 2 and Activity 3 is the point of running them back to back.
4. GenAI is excellent at populating bones fast (breadth at near-zero cost) and poor at knowing which are real. Note how many causes it returned as [HYPOTHESIS] — those are the ones your team must go and verify. The AI generates; the human evidences.

Self-check — is your output finished?

Your fishbone is complete when: every bone has at least three entries; every entry is marked [EVIDENCED] or [HYPOTHESIS]; every entry is phrased as a cause ('single queue with no triage') rather than a complaint ('the queue is terrible') or a solution ('add more counters'); and you have found at least one cross-bone arrow. If any bone is empty, you have a visibility gap, not an absence of causes — say so explicitly.

Activity 4 — Pareto Analysis to Target the Vital Few

Objective

A1, A3 · K2 — Prioritise causes by contribution and identify key implications.

At a glance

Field	Detail
Case study	Nexa Logistics — 2,000 failed deliveries a month
Duration	40 minutes
Grouping	Teams of 3-5
Tools	Pareto Chart ed-tool, ChatGPT / Copilot / Gemini
Ed-tool	Pareto Chart Tool — https://alfredang.github.io/paretochart/

The scenario

Nexa Logistics runs last-mile delivery for e-commerce clients across Singapore. Of 40,000 monthly deliveries, 2,000 fail on first attempt — a 5% failure rate against a contractual SLA of 2%. Every failed delivery costs S$8.50 to re-attempt: S$17,000 a month, and two clients have invoked SLA penalty clauses.

The Operations Director has a list of eleven failure reasons and a budget that will fund roughly two initiatives this quarter. His instinct is to start with 'Address incorrect' because it generates the angriest client emails.

Your team must decide where the two initiatives should go — using the data, not the noise.

The evidence

Nexa Logistics — first-attempt delivery failures by reason, last month

Failure reason	Count	Cost impact (S$)
Recipient not at home (no delivery window given)	742	6307
Driver could not access condo / office lobby	486	4131
Address incomplete or incorrect	231	1964
Parcel not loaded (sorting error)	188	1598
Recipient refused (wrong item expected)	121	1029
Vehicle breakdown / route disruption	94	799
Weather-related suspension	61	519
Recipient contact number	38	323

invalid		
Parcel damaged in transit	24	204
Restricted delivery hours at destination	20	170
Other / unclassified	15	128

Step-by-step

52. Read the Nexa Logistics scenario and the failure reason table.
53. Open the Pareto ed-tool at https://alfredang.github.io/paretochart/
54. Enter each failure reason and its count. The tool sorts descending and computes the cumulative line.
55. Read off the cumulative percentage and mark where the line crosses 80%.
56. Count how many reasons sit to the left of that line — these are your vital few.
57. Calculate by hand: a 60% reduction on the top two gives what new failure rate? Compare against the 2% SLA.
58. Run the GenAI Pareto prompt from the slide and check the AI's arithmetic against your own. Note any error.
59. Restate the top two reasons so that Nexa — not the customer — owns the fix.
60. Decide where the two funded initiatives should go and prepare a one-minute case for the Operations Director.
61. Present, including the arithmetic showing whether the SLA is reachable.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as an operations analyst. Perform a Pareto (80/20) analysis on the delivery
failure data below for a Singapore last-mile logistics company.

Total deliveries: 40,000/month. Failures: 2,000 (5%). SLA target: 2%. Re-
attempt cost: S$8.50 each.

Data (reason, count):
Recipient not at home 742; Lobby access denied 486; Address incorrect 231;
Parcel not loaded 188; Recipient refused 121; Vehicle breakdown 94; Weather 61;
Invalid contact 38; Damaged 24; Restricted hours 20; Other 15.

Produce:
1. A table with count, percentage, and CUMULATIVE percentage, sorted descending
2. The cut-off line — which reasons make up the vital few (roughly 80%)
3. The monthly S$ recovery if the top 2 reasons were reduced by 60%
4. Whether hitting the 2% SLA is achievable by fixing the vital few alone —
show the arithmetic
5. One caution about what this Pareto analysis does NOT tell me
```

Discussion questions

62. Build the Pareto chart. How many of the eleven reasons account for roughly 80% of failures? What does that ratio mean for how the Operations Director should spend his budget?
63. The Director wants to start with 'Address incorrect' because it generates the angriest emails. What does the data say, and how would you make that argument to him without dismissing his concern?
64. Calculate: if the top two reasons drop by 60%, what is the new failure rate? Does that meet the 2% SLA? Show your arithmetic.
65. The top two reasons — 'not at home' and 'lobby access' — look like customer problems. Reframe each as something Nexa controls. What changes about who owns the fix?
66. Pareto ranks by FREQUENCY. Name a situation in your own workplace where the most frequent problem is not the most important one. What would you rank by instead?

Debrief

EXPECTED ANALYSIS:
Not at home 742 (37.1%, cum 37.1%)
Lobby access 486 (24.3%, cum 61.4%)
Address incorrect 231 (11.6%, cum 73.0%)
Parcel not loaded 188 (9.4%, cum 82.4%) <- the 80% line falls here
The remaining 7 reasons together account for under 18%.

So FOUR of eleven reasons carry 82% of the failures — and the top TWO alone carry 61%.

ARITHMETIC: a 60% reduction on the top two removes 0.6 x 1,228 = 737 failures, leaving 1,263 of 40,000 = 3.2%. That is a large win but it does NOT reach the 2% SLA (which needs failures under 800). Reaching SLA requires the top two AND a meaningful dent in reasons 3 and 4. Learners who claim the SLA is met have not done the arithmetic — make them show it.

KEY TEACHING POINTS:
1. 'Address incorrect' is only 11.6% — third place. It is loud, not large. The Director is responding to escalation volume, which correlates with client temperament, not failure frequency. The Pareto chart is how you have that conversation with evidence rather than opinion. This is the transferable political skill in this activity.
2. REFRAMING OWNERSHIP is the deeper lesson: 'recipient not at home' is only a customer problem if you never offered a delivery window. 'Lobby access denied' is only a customer problem if you never negotiated building access protocols. Both become Nexa-controllable the moment you restate them — and 61% of the problem changes owner.
3. LIMITATION: Pareto ranks by frequency, and frequency is not severity. A single safety-critical or regulatory failure can outrank 700 inconveniences. Always ask what the right ranking dimension is — count, cost, risk or customer harm. Here, cost tracks count

closely (uniform S$8.50), which is why the two rankings agree; that is not always true.
4. Pareto tells you WHERE to look, never WHY. The 742 'not at home' failures still need a 5 Whys. The tools chain: Pareto for priority, then 5 Whys for cause.

Self-check — is your output finished?

Your analysis is sound when: the cumulative column reaches 100%; you can name the vital few and the exact cumulative percentage at your cut-off; you have shown the arithmetic on whether the 2% SLA is reachable (it is not, on the top two alone); and you can state in one sentence what Pareto does NOT tell you. Verify the AI's percentages against the tool's — if they disagree, the tool is right and you have just learned something about trusting AI arithmetic.

Activity 5 — System Loops: Why the Problem Keeps Coming Back

Objective

A3 · K2 — Deduce linkages and patterns to identify key implications on organisational systems.

At a glance

Field	Detail
Case study	Horizon Bank revisited — the fix that made it worse
Duration	50 minutes
Grouping	Teams of 3-5
Tools	System Loop ed-tool, ChatGPT / Copilot / Gemini
Ed-tool	System Loop Tool — https://alfredang.github.io/systemloop/

The scenario

Return to Horizon Bank Jurong East. Acting on the fishbone, the Regional Director did the obvious thing: he ordered two additional counters staffed at peak, redeploying staff from the back office.

Month 1: average wait fell from 58 to 34 minutes. Everyone celebrated.
Month 4: average wait is back to 61 minutes — WORSE than before. Complaints are at 68 per month. Three more tellers have resigned. Back-office processing is now two days behind, which has started generating a new complaint category of its own.

The Regional Director is baffled: "We added capacity and the problem came back bigger. What am I missing?"

What he is missing is that a branch is not a queue — it is a system of feedback loops. Your team must map them.

The evidence

Horizon Bank — system variables and observed 4-month movements

Variable	Month 1	Month 4	Direction
Counters staffed at peak	3.2	5.1	up
Average wait time (min)	58 -> 34	61	down then up
Back-office processing backlog (days)	0.5	2.0	up
Teller errors requiring rework	6%	14%	up
Rework returning to the counter	low	high	up
Teller overtime hours / month	88	196	up
Staff resignations (quarter)	2	3	up
Share of counter staff under 6 months	44%	58%	up
Mean transaction handling time (min)	9	13	up
Customer complaints / month	60	68	up
Digital channel adoption	21%	20%	flat

Step-by-step

67. Read the revisited Horizon Bank scenario and the 4-month variable movements.
68. Open the System Loop ed-tool at https://alfredang.github.io/systemloop/
69. List the key variables from the table as nodes: counters staffed, wait time, backlog, errors, rework, overtime, resignations, inexperience, handling time, digital adoption.
70. Draw the loop the Director INTENDED: pressure -> counters -> capacity -> wait time down. Label it B1 and confirm it closes as balancing.
71. Now trace the redeployment: where did the back-office staff come from, and what happened downstream? Close that loop and label it R1.
72. Trace the workload consequences on people: overtime, resignations, inexperience, errors. Close it and label it R2.
73. Mark (+) or (-) on every link. Verify each loop: an even number of (-) links makes it reinforcing; an odd number makes it balancing.
74. Identify where the DELAY sits — which loop acts in days and which acts in months.
75. Ask which balancing loop is available but not running. Add B2 (digital adoption) and mark it dormant.
76. Run the GenAI system-thinking prompt from the slide and compare its loops against yours.
77. Circle two leverage points where breaking ONE link would collapse a reinforcing loop.

78. Present: the four loops, why month 1 lied to you, and your two leverage points.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as a systems thinking facilitator. Help me map the causal feedback loops in
this case.

Situation: A bank branch added 2 counters at peak by redeploying back-office
staff. Wait time fell from 58 to 34 minutes in month 1, then rose to 61 minutes
by month 4 — worse than the starting point.

Observed changes over 4 months: back-office backlog 0.5 -> 2.0 days; teller
errors 6% -> 14%; rework returning to the counter rose; overtime 88 -> 196
hrs/month; resignations rose; share of tellers under 6 months 44% -> 58%; mean
handling time 9 -> 13 min; digital adoption flat at ~20%.

Produce:
1. At least 2 REINFORCING (R) loops and 2 BALANCING (B) loops. For each, list
the variables in causal order with (+) or (-) on each link, and name the loop
2. Explain specifically WHY the month-1 improvement reversed — which loop
overwhelmed which
3. Identify the delay in the system and explain why the delay made the wrong
decision look right
4. Name 2 leverage points that would break the reinforcing loops, and say what
makes them leverage rather than just more effort
```

Discussion questions

79. Map at least two reinforcing (R) and two balancing (B) loops. Which loop did the Regional Director believe he was pulling, and which loop did he actually trigger?
80. Month 1 showed real improvement. Explain the DELAY: why did the harmful loop take three months to overwhelm the helpful one, and what does that mean for how we evaluate quick wins?
81. Follow the back-office redeployment all the way round. Trace how a decision about counters ended up increasing teller resignations.
82. Digital adoption stayed flat at 20% throughout. Which loop is NOT running that should be, and why is a dormant balancing loop as dangerous as an active reinforcing one?
83. Name two leverage points. What makes a leverage point different from simply doing more of the same thing harder?

Debrief

EXPECTED LOOPS:

B1 — CAPACITY CONTROL (balancing, what the Director intended): wait time up (+) -> management pressure (+) -> counters staffed (+) -> service capacity (+) -> wait time

DOWN (-). This is the loop that produced the month-1 win. It is real, but it is not the only loop running.

R1 — REDEPLOYMENT BACKFIRE (reinforcing, the loop he triggered without seeing it): counters staffed (+) -> back-office staff removed (+) -> processing backlog (+) -> errors and unresolved cases (+) -> rework returns to the counter (+) -> counter workload (+) -> handling time (+) -> WAIT TIME (+) -> more pressure to staff counters (+). Handling time rising from 9 to 13 minutes is the fingerprint of this loop.

R2 — BURNOUT SPIRAL (reinforcing): workload (+) -> overtime (+) -> stress and burnout (+) -> resignations (+) -> share of inexperienced tellers (+) -> error rate (+) -> rework (+) -> workload (+). Tellers under 6 months rising 44% -> 58% while errors doubled 6% -> 14% is the fingerprint here.

B2 — DIGITAL MIGRATION (balancing, DORMANT): wait time (+) -> willingness to try digital (+) -> digital adoption (+) -> branch transaction volume (-) -> wait time (-). Adoption stayed flat at ~20%, so this loop never engaged. Nothing was done to activate it.

KEY TEACHING POINTS:
1. THE CENTRAL INSIGHT: he pulled B1 and unknowingly powered R1 and R2. Because R-loops compound, they eventually overwhelm a one-off capacity increase. Adding capacity to a system with an active reinforcing loop buys a temporary win and a bigger relapse.
2. DELAY IS THE TRAP: B1 acts in days (staff a counter, wait drops). R1 and R2 act over months (backlog builds, errors accumulate, people quit). The delay is precisely why the month-1 result validated a decision that was making things worse. Every learner should leave able to say: 'a quick win measured too early is indistinguishable from a mistake.' This is the direct bridge into Topic 3's leading vs lagging indicators.
3. DORMANT LOOPS: B2 was available the whole time and nobody activated it. Shifting simple transactions to digital reduces the load that feeds R1. Asking 'which balancing loop is not running?' is often more productive than adding effort to loops that are.
4. LEVERAGE vs EFFORT: more counters = effort (you must sustain it forever, and it feeds R1). Leverage = breaking a link in a reinforcing loop: triage complex cases away from the counter; eliminate the triple data entry so error rate falls; protect back-office capacity so rework never returns to the counter; activate B2 with in-branch digital ambassadors. Leverage changes the structure; effort fights the structure.
5. This is why Topic 1 ends here. 5 Whys gave depth, Fishbone gave breadth, Pareto gave priority — and System Loops explains why well-intentioned fixes rebound. Learners now have the full diagnostic set before touching solutions in Topic 2.

Self-check — is your output finished?

Your loop map is valid when: every loop CLOSES (the last variable feeds back to the first); every link carries a (+) or (-); each loop is correctly typed as R or B by counting negative links; you can explain the month-1-to-month-4 reversal by naming which loop overwhelmed which; and your leverage points break a LINK in a reinforcing loop rather than adding more effort to a balancing one. If your answer is 'hire more people', you have described effort, not leverage.

Activity 6 — Divergent Ideation with GenAI (Breadth Before Judgement)

Objective

A5 · K4 — Generate a wide solution set using appropriate problem-solving techniques.

At a glance

Field	Detail
Case study	Meridian Health — solving the retention root cause
Duration	45 minutes
Grouping	Teams of 3-5
Tools	ChatGPT / Copilot / Gemini, Pinboard
Ed-tool	Pinboard — https://alfredang.github.io/pinboard/

The scenario

Your 5 Whys in Activity 2 established the root cause at Meridian Health: a workforce retention and onboarding failure that leaves 40% of the 07:00 roster staffed by untrained relief phlebotomists at 14 minutes per draw instead of 6.

The Clinical Director has accepted the analysis and killed the S$1.2m seventh-centre proposal. She now wants options — and she has been explicit: "Don't bring me three sensible ideas. Bring me twenty, including the ones that sound ridiculous, and then tell me which four are real."

Budget guidance: up to S$250,000 over 12 months. Constraint: MOH clinical protocols on phlebotomy competency cannot be relaxed under any circumstances.

The evidence

Meridian Health — solution constraints and context

Constraint	Detail
Budget	Up to S$250,000 over 12 months
Clinical	MOH phlebotomy competency standards are non-negotiable
Contract	07:00-09:30 fasting draw window is contractual with corporate clients
Courier	Lab courier at 09:45 is fixed by the external lab partner
Workforce	Permanent phlebotomist turnover 44%/yr; relief pool is agency-supplied
Training	Current onboarding is 2 days shadowing, no structured sign-off

Market	Phlebotomist salaries are within 5% of the market median
Shift	Early shift (06:30 start) attracts no differential pay

Step-by-step

84. Restate the Meridian root cause from Activity 2 in one sentence so the whole team is solving the same thing.
85. MANUAL ROUND FIRST — set a 7-minute timer. Each person writes ideas silently on post-its. No discussion, no evaluation, no 'that won't work'.
86. Post all ideas to the wall at https://alfredang.github.io/pinboard/ and count them.
87. Now run the GenAI divergent ideation prompt from the slide.
88. Add every AI idea your team did not already have to the board, in a different colour.
89. Count the two groups. Discuss: which AI ideas would nobody in your team have proposed in a real meeting, and what stopped them?
90. Run the SCAMPER section of the prompt against the '07:00-09:30 at the centre' assumption.
91. For each SCAMPER letter, add the resulting idea to the board.
92. Go through the board and mark each idea HARD CONSTRAINT (genuinely ruled out by MOH or contract) or HABIT (only feels fixed).
93. Do NOT rank or cost anything yet — carry the full board forward to Activity 7.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as an innovation consultant running a divergent ideation session. Do NOT
evaluate or filter yet.

Root cause to solve: A Singapore health-screening operator has 44% annual
turnover among permanent phlebotomists. Vacancies are backfilled by agency
relief staff who take 14 min per blood draw versus 6 min for trained staff. 40%
of the 07:00 roster is relief staff, causing 31% of morning appointments to
overrun and 23% of samples to miss the fixed 09:45 lab courier.

Constraints: S$250k over 12 months; MOH phlebotomy competency standards cannot
be relaxed; the 07:00-09:30 fasting window and the 09:45 courier are both
fixed.

Generate 20 distinct solutions grouped under:
- People & Retention
- Training & Onboarding
- Process & Scheduling
- Technology & Automation
- Customer/Client Experience

Rules:
```

```
- Include at least 4 ideas that would normally be dismissed as unrealistic
- Include at least 2 ideas borrowed from a COMPLETELY different industry (say
which)
- One line each, phrased as an action
- Do NOT rank, cost or evaluate them

Then, separately: apply SCAMPER (Substitute, Combine, Adapt, Modify, Put to
another use, Eliminate, Reverse) to the single assumption that 'every draw must
be done by a phlebotomist at the centre between 07:00 and 09:30' and give one
idea per SCAMPER letter.
```

Discussion questions

94. Your team generated ideas manually first, then with GenAI. Compare the two lists — how many did the AI produce that your team would never have said out loud, and why not?
95. The AI was asked for two ideas borrowed from another industry. Which cross-industry idea is most transferable to Meridian, and what would have to be true for it to work?
96. Apply SCAMPER's 'Reverse' and 'Eliminate' to the fixed 07:00-09:30 window. Which supposedly fixed constraint turns out to be an assumption rather than a real constraint?
97. Identify one idea on your list that is genuinely ruled out by the MOH clinical constraint. How do you tell a hard constraint from a habit?
98. Cognitive fixedness is the tendency to reach for the familiar solution. Find one idea on your list that only appeared because you deliberately broke fixedness — what triggered it?

Debrief

EXPECTED RANGE — a strong list spans all five groups and includes at least these shapes:

PEOPLE & RETENTION — early-shift differential pay; career ladder to senior phlebotomist; convert best relief staff to permanent; retention bonus at 12 and 24 months; exit-interview programme to find the real reason people leave.

TRAINING & ONBOARDING — structured 10-day onboarding with competency sign-off; simulation arm practice before live draws; buddy system pairing relief with permanent staff; a micro-credential the agency pool can earn before their first shift.

PROCESS & SCHEDULING — stagger appointment slots so relief staff take simpler draws; roster all 5 permanent staff at 07:00 and taper later; pre-screen difficult-draw patients to trained staff; negotiate a second courier pickup.

TECHNOLOGY — vein-finder devices to cut the 18% repeat rate; digital pre-registration; automated fasting-compliance reminders the night before; queue app.

CLIENT EXPERIENCE — corporate on-site draws at the client's office; extend the window by agreement; tiered booking so the largest client gets trained staff.

KEY TEACHING POINTS:

1. DIVERGENCE MUST PRECEDE JUDGEMENT. Teams that evaluate while

generating typically stop at 6-8 ideas, all conventional. The instruction to include 'ridiculous' ideas exists because the ridiculous ones are where fixedness breaks — and the merely unusual ones next to them are often viable. Count your team's manual list against the AI list; the gap IS the lesson.
2. GENAI IS A DIVERGENCE ENGINE. This is its strongest contribution to problem solving: breadth at near-zero cost, free of the political self-censorship that stops a junior employee proposing 'pay the early shift more' in front of a Finance Director. The AI has no career risk. Note that it also has no judgement — which is Activity 7's job.
3. THE SCAMPER REVEAL: the biggest unlock is that the 07:00-09:30 window is fixed only because draws happen AT THE CENTRE. 'Reverse' (bring the draw to the client's office) and 'Eliminate' (remove the centre visit for corporate clients entirely) both dissolve the constraint rather than working within it. Meanwhile the MOH competency standard is a genuine hard constraint — it is a regulation, not a habit. Teaching learners to separate REGULATION from HABIT is the durable skill here.
4. CROSS-INDUSTRY TRANSFER: mobile phlebotomy mirrors mobile blood-donation drives; the buddy/competency-card model comes from aviation and F&B; surge differential pay comes from ride-hailing. Analogy is a recognised problem-solving strategy, and GenAI is unusually good at it because it has read across every domain.
5. Do NOT let teams rank yet. Ranking is Activity 7. Holding the line between divergence and convergence is itself the discipline being taught.

Self-check — is your output finished?

Your divergence is sufficient when: you have at least 20 distinct ideas spanning all five groups; at least 4 would raise an eyebrow in a management meeting; at least 2 are borrowed from another industry and you can name it; you have one idea per SCAMPER letter; and you have correctly separated at least one HARD CONSTRAINT from at least one HABIT. If every idea on your board is comfortable, you have not diverged — you have listed.

Activity 7 — Converging: Impact-Ease Matrix and Weighted Decision Matrix

Objective

A5 · K4, K6 — Shortlist and evaluate the most viable ideas using decision-making techniques.

At a glance

Field	Detail
Case study	Meridian Health — choosing four from twenty
Duration	50 minutes
Grouping	Teams of 3-5
Tools	ChatGPT / Copilot / Gemini, Pinboard
Ed-tool	Pinboard — https://alfredang.github.io/pinboard/

The scenario

You now hold twenty-plus candidate solutions from Activity 6. The Clinical Director will fund approximately four, inside S$250,000 over 12 months.

Two members of your team are already advocating loudly for their favourite ideas and the discussion is circling. This is exactly the moment where teams either make an evidence-based decision or default to whoever is most senior or most persistent.

Your job is to convert the board into a defensible shortlist using two complementary tools: a fast Impact-Ease screen, then a weighted decision matrix on the survivors.

The evidence

Meridian Health — agreed decision criteria and weightings

Criterion	Weight	Scoring guidance (1-5)
Impact on root cause (retention/training)	35%	5 = directly fixes the root cause
Speed to measurable effect	20%	5 = effect visible within 8 weeks
Cost within S$250k envelope	20%	5 = under S$25k
Clinical/regulatory risk	15%	5 = no MOH implication at all
Staff and client acceptance	10%	5 = actively welcomed by both

Step-by-step

99. Bring the full idea board from Activity 6. Remove any duplicates by merging them.
100. Draw the Impact-Ease matrix on a flip chart with four labelled quadrants.
101. Place every idea in a quadrant as a team. Where you disagree, place it on the line and move on — do not stall.
102. Discard the Thankless Tasks. Set the Fill-Ins aside as 'do if free capacity'.
103. BEFORE scoring anything, agree the five criteria weights as a team. Write them down. This prevents arguing about scores when you actually disagree about weights.
104. Build the weighted decision matrix for the surviving Quick Wins and Big Bets. Score each criterion 1-5.
105. Compute the weighted totals and rank.
106. Run the GenAI convergence prompt from the slide with your actual list pasted in.
107. Compare the AI ranking to yours. Investigate every place they disagree by more than two positions.
108. SENSITIVITY TEST: change the speed weight from 20% to 40% and re-rank. Record whether the winner changes.
109. Select your final FOUR as a PORTFOLIO — check they do not all attack the same cause, and check for redundancy.
110. Prepare a two-minute recommendation to the Clinical Director: the four, the total cost, the criteria, and the one thing you are still leaving unaddressed.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as a decision analyst. I will give you a list of candidate solutions. Do
NOT add new ideas.

STEP 1 — Classify every solution on an Impact vs Ease matrix as exactly one of:
- Quick Win (high impact, easy)
- Big Bet (high impact, hard)
- Fill-In (low impact, easy)
- Thankless Task (low impact, hard)
Give a one-line reason for each placement.

STEP 2 — Take only the Quick Wins and Big Bets and score them in a weighted
decision matrix:
- Impact on root cause (weight 35%)
- Speed to measurable effect (20%)
- Cost within a S$250k envelope (20%)
- Clinical/regulatory risk (15%)
- Staff and client acceptance (10%)
Score each 1-5, show the weighted total, and rank them.

STEP 3 — Recommend the best FOUR as a portfolio, and explicitly state:
- why this combination is stronger than the top four individual scores
- which two solutions overlap or make each other redundant
- what the portfolio still leaves unaddressed

Solutions: [paste your team's list here]
```

Discussion questions

111. Place all your ideas on the Impact-Ease matrix. Which quadrant is most crowded, and what does a crowded Fill-In quadrant tell you about how your team was thinking?
112. Run the weighted matrix. Did the top-scoring solution match your team's gut favourite from Activity 6? If not, what did the weighting expose?
113. Change the weight on 'Speed to measurable effect' from 20% to 40% and re-rank. Does the winner change? What does that sensitivity tell you about how robust your recommendation is?
114. The AI recommended a portfolio of four. Explain why the best four INDIVIDUAL scores are not automatically the best four TOGETHER.
115. Consensus fatigue is when a team accepts a weak option just to end the discussion. Where in this activity was your team most at risk of it, and what stopped you?

Debrief

EXPECTED SHAPE OF A STRONG ANSWER:
QUICK WINS — roster all 5 permanent phlebotomists at 07:00 and taper later (near-zero cost, immediate effect); early-shift differential pay; pre-visit fasting-compliance

reminders; route difficult draws to trained staff.

BIG BETS — structured 10-day onboarding with competency sign-off; convert top relief staff to permanent; corporate on-site draws.

FILL-INS — better signage, updated leaflets, minor booking-form changes.

THANKLESS — a new centre; a full HR system replacement.

A defensible portfolio of four usually pairs an immediate stabiliser with a root-cause fix, e.g.: (1) re-roster the 07:00 shift — buys relief NOW at almost no cost; (2) early-shift differential — attacks why people leave the early shift; (3) structured onboarding with competency sign-off — attacks the 14-min relief draw time and the 18% repeat rate; (4) convert top relief staff to permanent — converts the agency pool into retained capacity.

KEY TEACHING POINTS:
1. TWO TOOLS, TWO JOBS. Impact-Ease is a fast screen that clears the board of Fill-Ins and Thankless Tasks in minutes. The weighted matrix is the rigorous instrument you use on the survivors and, crucially, the one you can DEFEND to a board. Using the heavy tool on all twenty wastes the session; using only the light tool leaves you unable to justify the call.
2. THE WEIGHTS ARE THE REAL DECISION. Most teams argue about scores when the disagreement is actually about weights. Surfacing weights first — before any scoring — converts a political argument into an explicit, recorded choice. This is the single most useful facilitation move in the activity.
3. SENSITIVITY TESTING: if doubling the speed weight flips the winner, your recommendation is fragile and you must say so. If the same solution wins under several weightings, you have a robust recommendation. Boards trust the second kind. Teach learners to run the test BEFORE they present, not after they are challenged.
4. PORTFOLIO, NOT LEADERBOARD: the top four scores may all attack the same cause and leave another exposed, or two may be redundant (on-site corporate draws largely removes the need for the 07:00 re-roster for those clients). A portfolio balances quick stabilisation against durable root-cause repair. This is where human judgement outperforms the AI's ranking, and learners should see that clearly.
5. AI LIMITATION TO NAME OUT LOUD: the AI will confidently invent cost figures and timelines it has no basis for. Its ranking is only as good as the criteria you set. It is a fast, tireless, unbiased-by-politics scorer — not a decision maker. The Clinical Director will ask YOU why, not the model.

Self-check — is your output finished?

Your shortlist is defensible when: every surviving idea sits in a named quadrant; the weighted matrix shows criteria, weights, scores and totals; you have run at least one sensitivity test and can state whether the winner held; your four are a portfolio rather than the top four scores; the total cost fits inside S$250k; and you can name one thing the portfolio does NOT fix. If you cannot explain to the Clinical Director why idea #5 lost, the matrix has not done its job.

Activity 8 — Building the Corrective Action Plan

Objective

A4 · K9 — Develop corrective action plans for shortfalls identified.

At a glance

Field	Detail
Case study	Meridian Health — from chosen solutions to an auditable plan
Duration	45 minutes
Grouping	Teams of 3-5
Tools	ChatGPT / Copilot / Gemini, Pinboard
Ed-tool	Pinboard — https://alfredang.github.io/pinboard/

The scenario

The Clinical Director has approved your four solutions. She now asks the question that separates a good analysis from a delivered result:

"Who is doing what, by when, with what money, and how will I know it worked?"

She also adds a warning from experience: "The last improvement project we ran had a beautiful slide deck and no owner. Six months later nothing had changed and everyone assumed someone else was doing it."

Your team must convert four approved solutions into a corrective action plan that would survive an internal audit — and that a colleague could execute without you in the room.

The evidence

Corrective Action Plan — mandatory components (K9)

Component	Why it exists	Failure if missing
Root cause addressed	Ties the action to the diagnosis	Solution drifts to a symptom
Corrective action	The specific thing being done	Vague intent, no execution
Owner (named person)	Single point of accountability	Everyone assumes someone else
Timeline / milestone	Makes progress checkable	Slips indefinitely, unnoticed
Resources required	Budget, people, tools committed	Stalls at first resource conflict
Success measure + baseline	Defines what 'worked' means	Cannot evaluate; endless debate
Risk and mitigation	Anticipates what breaks it	Surprised by predictable failure
Review checkpoint	Forces a decision to continue/stop	Runs on past the point of failure

Step-by-step

116. List your four approved solutions from Activity 7 across the top of a flip chart.

117. Draw the eight CAP components down the side as rows.
118. Fill in 'Root cause addressed' for all four FIRST. If any solution cannot be traced to your diagnosed root cause, challenge whether it belongs in the plan.
119. Assign a NAMED ROLE as owner for each action. Reject any department name. One owner per action — no co-owners.
120. Add timelines with a start, a milestone and a completion date. Use week numbers, not 'Q3'.
121. Cost each action and check the four together fit inside S$250,000.
122. Write each success measure as metric + baseline + target. Cross-check the baseline against your Activity 1 problem statement.
123. Test each measure: can Meridian measure this with data it already collects? If not, either change the measure or add 'establish measurement' as a task.
124. Add the key risk and a mitigation that is itself an action with an owner — not a hope.
125. Set a review checkpoint for each action and state the DECISION to be made at it.
126. Run the GenAI corrective action plan prompt from the slide.
127. Act on the AI's three flags: sequence the dependent actions, rebalance any overloaded owner, and fix any unmeasurable success measure.
128. Post the completed plan to https://alfredang.github.io/pinboard/ and present one row in full to the room.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
You are an execution-focused planning assistant. Convert each approved solution
into a corrective action plan row. Be concrete and brief — no theory.

For EACH solution, produce exactly these fields:
- Root cause addressed (link back to the diagnosis)
- Corrective action (one specific sentence)
- Owner (a job title, not a department)
- Timeline (start, key milestone, completion)
- Resources (budget in S$, people, tools)
- Success measure (metric + baseline + target)
- Key risk + mitigation
- Review checkpoint (date and the decision to be made at it)

Then flag, in a separate section:
1. Any two actions that depend on each other and must be sequenced
2. Any owner who appears on more than two actions (overload risk)
3. Any success measure that cannot actually be measured with data the
organisation already has

Approved solutions: [paste your four here]
```

```
Root cause: 44% phlebotomist turnover leaving 40% of the 07:00 roster
untrained, causing 14-min draws vs 6-min, 31% appointment overruns and 23% of
samples missing the 09:45 courier.
Budget: S$250,000 over 12 months.
```

Discussion questions

129. Complete all eight components for each of your four actions. Which component did your team find hardest to fill honestly, and what does that reveal?
130. Every owner must be a named role, not a department. Why does 'Operations' fail as an owner where 'Centre Operations Manager' succeeds?
131. Check your success measures against the baseline in your original problem statement. Can each one actually be measured with data Meridian already collects? Which cannot?
132. The AI flagged dependencies between actions. Which two of your actions must be sequenced, and what breaks if you run them in parallel?
133. Identify the biggest risk across your whole plan. Is your mitigation a real action with an owner, or a hope?

Debrief

WORKED EXAMPLE — one complete row to standard:
Root cause addressed: Untrained relief staff on the 07:00 roster (14-min vs 6-min draws).
Corrective action: Implement a structured 10-day phlebotomy onboarding with formal competency sign-off before any unsupervised draw.
Owner: Clinical Training Lead.
Timeline: Design by week 3; pilot at two centres weeks 4-8; full rollout by week 14.
Resources: S$65,000 (trainer time, simulation arms, backfill cover); 0.4 FTE for 14 weeks.
Success measure: Mean relief-staff draw time from 14 min (baseline) to under 8 min by week 16; first-attempt success from 82% to 92%.
Key risk: Backfill cover unavailable during pilot, so centres refuse to release staff. Mitigation: Clinical Director pre-commits agency budget for pilot weeks; escalation path named.
Review checkpoint: Week 8 — decision to roll out, extend the pilot, or stop.

KEY TEACHING POINTS:
1. THE OWNER IS THE PLAN'S SPINE. 'Operations' cannot be phoned, cannot be held to a date and cannot be asked why it slipped. A named role can. The Clinical Director's warning in the scenario is the most common real-world failure of improvement projects, and it is entirely preventable with one discipline: one named owner per action, no exceptions, no co-owners.
2. THE MEASURE MUST TIE TO THE BASELINE FROM TOPIC 1. This is where the course closes its own loop: the problem statement's baseline in Activity 1 is what makes evaluation possible in Topic 3. A team that skipped the baseline now literally cannot write this field — let them discover that themselves; it lands harder than being told.

3. MEASURABILITY CHECK: 'improved staff morale' is not measurable with data Meridian holds. 'Voluntary resignations per quarter' and 'early-shift roster fill rate' are. Force the substitution now, not at review time.
4. SEQUENCING: converting relief staff to permanent DEPENDS on the onboarding programme existing — convert first and you have permanent staff trained to the old inadequate standard, locking in the defect. Dependencies are where good plans quietly fail.
5. OWNER OVERLOAD: if the Centre Operations Manager owns three of four actions, the plan is a single point of failure regardless of how good each row looks. GenAI is genuinely useful at spotting this pattern mechanically — a good example of AI catching what a team misses because they are too close to it.
6. THE REVIEW CHECKPOINT MUST CARRY A DECISION. 'Review progress' is theatre. 'Decide to roll out, extend or stop' is governance.

Self-check — is your output finished?

Your CAP is audit-ready when: all eight components are complete for all four actions; every owner is a named role and no role owns more than two actions; every success measure has a metric, a baseline number and a target number; every measure is obtainable from data the organisation already has (or has an explicit task to start collecting it); dependencies are sequenced; every mitigation is an action with an owner; and every checkpoint names a decision. The real test: hand the plan to another team and ask them to execute row one without asking you a single question.

Activity 9 — Implementation Planning and Stakeholder Resistance

Objective

A6 · K3, K8 — Draw up implementation plans and address factors affecting effectiveness.

At a glance

Field	Detail
Case study	ShopFront SG — the rollout that has to survive contact with people
Duration	50 minutes
Grouping	Teams of 3-5
Tools	ChatGPT / Copilot / Gemini, Pinboard
Ed-tool	Pinboard — https://alfredang.github.io/pinboard/

The scenario

Return to ShopFront SG from Activity 1. The diagnosis is complete and the solution set is approved: a one-page guest checkout, a CDN and caching layer to bring app load under 2 seconds, a supplier QC checklist to cut the return rate, and cart-recovery notifications.

The Chief Operating Officer has seen improvement projects die before. She says:

"The plan is fine. Now tell me who is going to fight it, why, and what you are going to do about it. Last year we bought a S$400,000 CRM that nobody used. It worked perfectly. That was not the problem."

Your team must build the implementation plan AND the stakeholder strategy that makes it stick. Assume nothing about goodwill.

The evidence

ShopFront SG — stakeholder map and known positions

Stakeholder	Interest	Likely position
IT Development team	Already at capacity on a POS migration	Resist — sees added workload
Head of Marketing	Sponsored the S$180k discount campaign	Resist — solution implies her spend was wrong
Store Operations	In-store sales flat, feels unaffected	Indifferent — 'not my problem'
Suppliers (12 vendors)	Face new QC standards and possible delisting	Resist — cost and scrutiny
Customer Service	Handling 410 tickets/week, overloaded	Support — expects relief
Finance	Approved budget, wants ROI evidence	Conditional — needs measurement
COO (sponsor)	Accountable for the turnaround	Champion
Frontline retail staff	Fear digital shift reduces store headcount	Anxious — job security

Step-by-step

134. Review the four approved ShopFront SG solutions and the stakeholder map.
135. Build the implementation plan: for each solution write Action, Method/Tool, Owner, Timeline.
136. Draw the four solutions on a timeline and mark dependencies with arrows. Identify what must not run in parallel.
137. For each of the eight stakeholders, write their objection IN THEIR OWN WORDS — first person, as they would actually say it in a meeting.
138. For each, decide what they need to hear or receive to move from their current position.
139. Assign a MESSENGER to each stakeholder and justify why that person rather than you.
140. Run the GenAI change management prompt from the slide.

141. Compare the AI's stakeholder analysis to yours. Note any objection it named more bluntly than your team was willing to write down — and ask why you softened it.
142. Rank all stakeholders by their power to block the rollout. Debate the top one as a team.
143. Build the 90-day communication plan: what, to whom, how often, by whom.
144. Post your stakeholder strategy to https://alfredang.github.io/pinboard/ and present your most dangerous stakeholder plus your mitigation.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as a change management and implementation planning advisor for a Singapore
fashion retailer.

Approved solutions: (1) one-page guest checkout; (2) CDN + caching to bring app
load from 4.8s to under 2.0s; (3) supplier QC checklist to cut the 12% return
rate; (4) automated cart-recovery notifications.

PART A — Implementation plan. For each solution give ONLY:
- Action (what will be done)
- Method/tool (how)
- Owner and timeline (who, by when)
- Sequencing note (what must happen before it)

PART B — Stakeholder strategy. Stakeholders and positions:
IT Development (at capacity, resists workload); Head of Marketing (sponsored a
S$180k campaign the diagnosis implies was misdirected — resists); Store
Operations (indifferent); 12 suppliers (face new QC standards — resist);
Customer Service (supportive, overloaded); Finance (conditional on ROI
evidence); COO (champion); frontline retail staff (fear job loss).

For EACH stakeholder give:
- Their specific objection in THEIR words, not yours
- What they need to hear or receive to move
- One concrete action to secure their cooperation
- Who is best placed to deliver that message and why

PART C — Name the single stakeholder most likely to sink this rollout, and
explain what makes them dangerous. Then give a 90-day communication plan: what
is communicated, to whom, how often, by whom.

Be blunt about political realities. Do not assume goodwill.
```

Discussion questions

145. Build the implementation plan for all four solutions. Which two MUST be sequenced rather than run in parallel, and what breaks if you ignore that?

146. The Head of Marketing resists because the diagnosis implies her S$180k campaign was misdirected. This is a face problem, not a logic problem. How do you secure her cooperation without requiring her to admit error?
147. Frontline retail staff fear the digital push costs them jobs. Is that fear irrational? What would you actually commit to — and what happens to your credibility if you promise something you cannot guarantee?
148. Twelve suppliers face new QC standards. What is the difference between imposing the checklist and getting them to adopt it, and which one survives after month three?
149. Identify the single stakeholder most likely to sink this rollout. Note: it may not be the loudest one. Justify your choice.

Debrief

EXPECTED IMPLEMENTATION SEQUENCING:
Guest checkout (weeks 1-4, quick win — builds momentum and credibility with Finance).
CDN/caching (weeks 2-10, big bet — MUST precede or accompany cart-recovery notifications; driving traffic back to a 4.8-second app converts a recovery campaign into a second bad experience and burns the customer twice).
Supplier QC (weeks 3-16, longest lead time — external parties, contractual change).
Cart recovery (weeks 10-14, AFTER load time is fixed).

STAKEHOLDER STRATEGY — the key moves:
MARKETING is the classic trap. The objection is not logical, it is reputational. The move is to reframe rather than relitigate: the campaign proved demand exists; conversion is where it leaks. Give her ownership of the cart-recovery workstream so she wins visibly from the fix. Delivered by the COO, peer to peer — never by the analyst who produced the diagnosis.
IT DEVELOPMENT resists capacity, not the idea. The move is real relief, not encouragement: re-sequence the POS migration, or fund contract capacity. Delivered by the COO because only the sponsor can re-prioritise.
SUPPLIERS respond to incentives, not instructions. Co-design the checklist with the three best-performing vendors, then tie compliance to order volume. An imposed checklist is complied with on paper for two months; a co-designed one with commercial consequence sticks.
FRONTLINE STAFF have a rational fear and deserve an honest answer. Commit to what you can actually control — no headcount reduction tied to this programme, and involvement in click-and-collect roles. Never promise what you cannot guarantee; one broken promise here costs you every subsequent change programme.
STORE OPERATIONS indifference is the quiet risk — click-and-collect needs them and they have no reason to care yet. Give them a stake before you need them.

MOST DANGEROUS STAKEHOLDER — accept a well-argued case for either, but push teams past the obvious: IT Development can hard-block delivery (nothing ships without them), while Marketing can soft-block through political attrition, which is harder to see and harder to escalate. Many teams name the loudest resister; the more sophisticated answer notes that INDIFFERENCE (Store Operations) often kills rollouts more quietly than opposition, because nobody escalates an absence of enthusiasm.

KEY TEACHING POINTS:
1. THE COO'S LINE IS THE WHOLE LESSON: the CRM 'worked perfectly. That was not the problem.' Solutions fail on ADOPTION far more often than on design. K8 — factors affecting the effectiveness of an implementation plan — is mostly about people, not technology.
2. MESSENGER MATTERS AS MUCH AS MESSAGE. The same words land differently from the COO, a peer, or the analyst. Teams consistently under-think this; make them assign a messenger to every message and justify it.
3. RESISTANCE IS INFORMATION. IT's objection reveals a genuine capacity conflict your plan had not costed. Suppliers' objection reveals the checklist needs commercial teeth. Treating resistance as data improves the plan; treating it as obstruction guarantees failure.
4. GenAI is unusually good here because it has no stake in the office politics — it will state plainly that Marketing's objection is about face, which a junior analyst may not dare write down. It is weak on the specific human relationships and history, which only the team knows. Divide the labour accordingly.

Self-check — is your output finished?

Your implementation plan is realistic when: every solution has an action, method, owner and timeline; dependencies are sequenced (cart recovery AFTER load time is fixed); every stakeholder has an objection written in their own voice, a specific move, and a named messenger; you have identified the highest-risk blocker with justification; and your communication plan runs the full 90 days rather than stopping at launch. If your plan assumes everyone cooperates because the analysis is correct, you have written a wish, not a plan.

Activity 10 — Measuring Effectiveness: Did It Actually Work?

Objective

A7 · K7 — Evaluate the effectiveness of implemented solutions and implementation plans.

At a glance

Field	Detail
Case study	ShopFront SG — 90 days later, and the numbers are ambiguous
Duration	45 minutes
Grouping	Teams of 3-5
Tools	ChatGPT / Copilot / Gemini, Pareto Chart ed-tool

Ed-tool	Pareto Chart Tool — https://alfredang.github.io/paretochart/

The scenario

Ninety days have passed. The COO has the results and she is not sure what to make of them. Some numbers moved well. Some did not move at all. One moved in the wrong direction. And the quarter included the Great Singapore Sale, which everyone is now using to explain whichever result suits their argument.

The Head of Marketing says the improvement is obviously seasonal. The IT lead says the improvement is obviously the CDN. Finance wants to know whether to release the next S$200,000 tranche.

Your team must evaluate honestly: what worked, what did not, what you cannot yet tell, and what you recommend next. "It's working" is not an acceptable answer.

The evidence

ShopFront SG — 90-day results against baseline

Metric	Baseline	Target	Day 90 actual	Verdict
App page load	4.8 s	under 2.0 s	1.7 s	Target met
Checkout completion	51%	68%	63%	Improved, short
Cart abandonment	74%	60%	64%	Improved, short
Online sales / month	S$1.9m	S$2.8m	S$2.5m	Improved, short
CSAT	70%	85%	72%	Barely moved
Complaint tickets / week	410	150	295	Improved, short
Product return rate	12%	5%	13%	WORSE
Repeat purchase (90-day)	29%	40%	31%	Barely moved
Guest checkout adoption	n/a	n/a	44% of orders	New data
Supplier QC checklist adoption	0	12 of 12	4 of 12	Behind
Cart recovery emails sent	n/a	n/a	18,400	Live
Cart recovery conversion	n/a	8%	3.1%	Underperforming

Step-by-step

150. Review the 90-day results table against the original problem statement baselines from Activity 1.

151. Assign a verdict to every metric: Met / Partially met / Not met / Deteriorated.
152. For each improved metric, ask: what CAUSED this? Mark each as Attributable, Partially attributable, or Confounded.
153. Investigate the return rate deterioration. Note that only 4 of 12 suppliers adopted the checklist — what does that tell you about whether the solution was ever actually tested?
154. Separate all metrics into LEADING and LAGGING. Check whether any target was set on an unrealistic horizon.
155. Identify how the Great Singapore Sale confounds the sales figures. Decide what analysis would isolate the programme effect.
156. Diagnose the cart recovery underperformance: wrong solution, wrong implementation, or wrong target? State what data would settle it.
157. Run the GenAI evaluation prompt from the slide.
158. Check the AI's attribution claims critically. Mark anywhere it claimed credit the evidence cannot support.
159. Trace the supplier QC shortfall back to your Activity 9 stakeholder analysis. Did you predict this resistance? What would you have done differently?
160. Write your recommendation to the CFO in three sentences: the verdict, the decision on the S$200k, and what would change your mind.
161. List the three things you would measure differently if you started again.

The GenAI prompt

Copy this prompt into your generative AI tool. Replace anything in <<double angle brackets>> with your own details.

```
Act as a performance evaluation analyst. Assess whether this improvement
programme worked. Be rigorous and do not flatter the results.

Results after 90 days (metric: baseline -> target -> actual):
App load 4.8s -> <2.0s -> 1.7s
Checkout completion 51% -> 68% -> 63%
Cart abandonment 74% -> 60% -> 64%
Online sales S$1.9m -> S$2.8m -> S$2.5m
CSAT 70% -> 85% -> 72%
Complaints/week 410 -> 150 -> 295
Product return rate 12% -> 5% -> 13% (WORSE)
Repeat purchase 29% -> 40% -> 31%
Supplier QC checklist adopted by 4 of 12 suppliers
Cart recovery conversion 3.1% against an 8% target

Context: the 90-day window included the Great Singapore Sale.

Produce:
1. A verdict per metric: Met / Partially met / Not met / Deteriorated
2. Which solutions can be credited with which movements — and where you CANNOT
establish attribution, say so explicitly and explain why
3. Why the product return rate got WORSE despite the QC initiative — give the
most likely explanation and state what evidence would confirm it
```

```
4. Separate LEADING from LAGGING indicators here, and explain why CSAT and
repeat purchase barely moved even though operational metrics improved
5. How the Great Singapore Sale confounds this evaluation, and what analysis
would isolate the programme effect from the seasonal effect
6. A clear recommendation to the CFO on releasing the next S$200k tranche:
continue, adjust, or stop — with reasoning
7. The three things you would measure differently next time
```

Discussion questions

162. Give a verdict on each metric. Overall — did this programme succeed? Defend a single-sentence answer to the COO.
163. The product return rate got WORSE (12% to 13%) despite the QC initiative. Give the most likely explanation, and say what evidence would confirm or refute it.
164. App load smashed its target but CSAT barely moved. Explain this using leading versus lagging indicators. How long would you expect CSAT to lag?
165. The Great Singapore Sale sits inside the measurement window. How would you isolate the programme effect from the seasonal effect? What would you have done differently at day 0 to make this answerable?
166. Cart recovery converts at 3.1% against an 8% target. Is the solution wrong, or the implementation, or the target? How would you tell the difference — and what does each answer imply?
167. Make the call to the CFO: release the next S$200,000, adjust, or stop. Justify it with the evidence, and name what would change your mind.

Debrief

EXPECTED EVALUATION:

ATTRIBUTION IS THE CENTRAL SKILL. App load 4.8s -> 1.7s is cleanly attributable to the CDN and caching — technical, direct, no plausible alternative cause. Checkout completion is PARTIALLY attributable to guest checkout (44% adoption is strong supporting evidence). Online sales, however, are CONFOUNDED by the Great Singapore Sale and cannot be cleanly attributed either way. A team that credits the full S$0.6m to the programme is overclaiming; a team that dismisses it as seasonal is underclaiming. The honest answer is: not determinable with this design, and here is what would have made it determinable.

THE RETURN RATE PUZZLE — the most instructive result on the board. Most likely explanation: only 4 of 12 suppliers adopted the QC checklist, so the intervention barely happened. Simultaneously, higher conversion and GSS volume brought in more first-time and discount-driven buyers, who return at a higher rate than loyal customers. So the return rate rose because of a MIX SHIFT, not because quality fell. Confirming evidence: segment return rate by supplier (adopters vs non-adopters) and by customer type (new vs repeat, full-price vs discounted). This is the moment learners see that a metric moving the wrong way does not automatically mean the solution failed — you must decompose before you conclude.

LEADING vs LAGGING: app load, checkout completion and guest checkout adoption are LEADING — they respond within days. CSAT, repeat purchase and brand perception are LAGGING — they reflect accumulated experience across multiple purchase cycles and typically lag two to three cycles, i.e. 6-9 months for a fashion retailer. Expecting CSAT to move in 90 days was a planning error, not an execution failure. The target itself was wrong.

CART RECOVERY at 3.1% vs 8%: distinguish the three diagnoses. Wrong solution (cart recovery does not work for this audience); wrong implementation (poor timing, weak copy, no incentive, emails hitting spam); wrong target (8% was benchmarked from a different sector). The way to tell: check open and click rates. High open + low conversion = offer/landing problem. Low open = deliverability or timing problem. Industry benchmarks for fashion cart-recovery sit around 3-5%, which strongly suggests the TARGET was unrealistic. Teams that immediately conclude 'the solution failed' have skipped the diagnosis — and that is exactly the behaviour the whole course exists to correct.

RECOMMENDATION TO THE CFO — the defensible answer is CONTINUE WITH ADJUSTMENT: the technical solutions delivered and are attributable; supplier QC did not actually get implemented (4 of 12) so it has not been tested and should not be abandoned on non-adoption; cart recovery needs its target rebased and its execution diagnosed; CSAT needs a longer horizon. Redirect effort to supplier adoption — which is a CHANGE MANAGEMENT failure, traceable straight back to Activity 9's prediction that suppliers would resist. That connection is the strongest single moment in the course: the resistance they predicted is the resistance that actually materialised, and it is the reason a metric missed.

KEY TEACHING POINTS:
1. 'IT'S WORKING' IS NOT AN EVALUATION. Metric-by-metric verdicts against baseline and target, with explicit attribution and explicit uncertainty, is.
2. EVALUATION IS ONLY POSSIBLE BECAUSE ACTIVITY 1 CAPTURED A BASELINE. Say this out loud — the course has now closed its loop, and learners should feel why the six-element problem statement mattered 16 hours ago.
3. DISTINGUISH SOLUTION FAILURE FROM IMPLEMENTATION FAILURE. Supplier QC did not fail; it was not adopted. Those demand completely different responses — one means change the solution, the other means fix the rollout.
4. CONFOUNDING IS ALWAYS PRESENT IN REAL WORKPLACES. You will rarely get a clean experiment. Good practice: define the measurement window at day 0, hold a control (e.g. staged rollout by region), and segment the data. Say what you cannot determine — that honesty is what makes the rest of your evaluation credible to a CFO.
5. GenAI will produce a confident, well-structured evaluation that quietly overclaims attribution unless you explicitly instruct it to flag what cannot be determined. Note that the prompt on the slide does exactly that — and compare it against what the AI returns without that instruction. That contrast is the AI-literacy takeaway of the entire course.

Self-check — is your output finished?

Your evaluation is credible when: every metric has a verdict against its baseline AND target; attribution is stated explicitly and confounded results are named as confounded rather than claimed; you can explain the worsening return rate without concluding the solution failed; you distinguish solution failure from implementation failure for supplier QC; your CFO recommendation names what would change your mind; and you can trace at least one missed metric back to a stakeholder resistance you predicted in Activity 9. If your evaluation credits the programme with everything that improved and blames the season for everything that did not, you have written advocacy, not evaluation.

The Prompt Library

These are the ten prompts used across the course, collected for reuse at work. Each is written to be pasted directly into ChatGPT, Microsoft Copilot, Google Gemini or Claude. Replace anything in <<double angle brackets>> with your own situation, and always paste your real evidence — without it the model will invent plausible substitutes.

Prompt 1 — Rewriting a Vague Problem into a Measurable Problem Statement

Use for: A1 · K1 — Identify the type of performance deficiency and express it as a measurable problem statement.

```
You are a business analyst for a Singapore fashion retailer.

Refine the vague statement below into ONE clear, measurable problem statement.

Vague statement: "Customers are unhappy and sales are dropping."

Evidence:
- App page load rose from 2.1s to 4.8s (benchmark <2.0s)
- Checkout completion fell 68% -> 51% (benchmark 70%)
- Cart abandonment rose 61% -> 74%
- CSAT fell 84% -> 70%
- Online sales fell S$2.8m -> S$1.9m per month; in-store flat

Your problem statement MUST contain, each labelled:
- Context (who, where, which process)
- Metric (the single primary measure)
- Baseline (where it is now, with the number)
- Target (where it must be, with the number)
- Timeframe (by when)
- Constraints (budget, headcount, compliance, brand)

Then list any assumption you had to make, and state what additional evidence
would let you remove that assumption. Do not propose solutions yet.
```

Prompt 2 — 5 Whys Root Cause Analysis

Use for: A2 · K2, K5 — Identify root causes with team members using structured group facilitation.

```
Act as a root cause analysis expert facilitating a clinical operations team.

Problem: 31% of morning health-screening appointments at Meridian Health
overrun by more than 40 minutes, and 23% of blood samples miss the fixed 09:45
lab courier.

Evidence: only 3 of 5 phlebotomists are rostered at 07:00 (all 5 by 08:30);
relief staff take 14 min per draw vs 6 min for trained staff; relief staff are
40% of the morning roster; 18% of draws need a repeat attempt; permanent staff
turnover is 44% a year.

Apply the 5 Whys technique. Ask ONE why at a time and wait for my answer before
asking the next. Go five levels deep.

At each level, state what EVIDENCE would confirm or refute that level — do not
accept my answer at face value. If my answer is an opinion rather than a fact,
say so and ask me for the data that would support it.

After level 5, summarise: the root cause, the evidence supporting it, and which
levels remain unverified assumptions.
```

Prompt 3 — Fishbone (Ishikawa) Diagram for Breadth of Causes

Use for: A2 · K2, K5 — Categorise potential causes across all contributing dimensions.

```
Act as a quality improvement facilitator running an Ishikawa (fishbone)
analysis with a bank branch team.

Problem (the fish head): Customer complaints at the Horizon Bank Jurong East
branch have risen to 60 per month (network average 14), with peak counter wait
time at 58 minutes against a 15-minute standard.

Generate candidate causes under these six bones: People, Process, Technology,
Material/Information, Environment, Measurement.

Rules:
- Give 4-6 candidate causes per bone, each phrased as a CAUSE not a complaint
- Mark each cause [EVIDENCED] or [HYPOTHESIS] based only on what I give you
below
- Do NOT propose solutions
- Finish by naming the 3 causes you would investigate first and say why

Evidence available: single queue with no triage; 61% of footfall between 11:30-
14:00; 3.2 of 6 counters staffed at peak; new KYC platform requires triple data
entry; screen response degraded 2s to 8s; 44% of tellers under 6 months tenure;
teller KPI is transactions-per-hour only; 34% of customers arrive without
documents; app checklist and branch signage disagree; complex cases take 27
```

```
minutes at the counter; no branch-to-HO feedback route; 18 seats in the waiting
area.
```

Prompt 4 — Pareto Analysis to Target the Vital Few

Use for: A1, A3 · K2 — Prioritise causes by contribution and identify key implications.

```
Act as an operations analyst. Perform a Pareto (80/20) analysis on the delivery
failure data below for a Singapore last-mile logistics company.

Total deliveries: 40,000/month. Failures: 2,000 (5%). SLA target: 2%. Re-
attempt cost: S$8.50 each.

Data (reason, count):
Recipient not at home 742; Lobby access denied 486; Address incorrect 231;
Parcel not loaded 188; Recipient refused 121; Vehicle breakdown 94; Weather 61;
Invalid contact 38; Damaged 24; Restricted hours 20; Other 15.

Produce:
1. A table with count, percentage, and CUMULATIVE percentage, sorted descending
2. The cut-off line — which reasons make up the vital few (roughly 80%)
3. The monthly S$ recovery if the top 2 reasons were reduced by 60%
4. Whether hitting the 2% SLA is achievable by fixing the vital few alone —
show the arithmetic
5. One caution about what this Pareto analysis does NOT tell me
```

Prompt 5 — System Loops: Why the Problem Keeps Coming Back

Use for: A3 · K2 — Deduce linkages and patterns to identify key implications on organisational systems.

```
Act as a systems thinking facilitator. Help me map the causal feedback loops in
this case.

Situation: A bank branch added 2 counters at peak by redeploying back-office
staff. Wait time fell from 58 to 34 minutes in month 1, then rose to 61 minutes
by month 4 — worse than the starting point.

Observed changes over 4 months: back-office backlog 0.5 -> 2.0 days; teller
errors 6% -> 14%; rework returning to the counter rose; overtime 88 -> 196
hrs/month; resignations rose; share of tellers under 6 months 44% -> 58%; mean
handling time 9 -> 13 min; digital adoption flat at ~20%.

Produce:
1. At least 2 REINFORCING (R) loops and 2 BALANCING (B) loops. For each, list
the variables in causal order with (+) or (-) on each link, and name the loop
2. Explain specifically WHY the month-1 improvement reversed — which loop
overwhelmed which
3. Identify the delay in the system and explain why the delay made the wrong
decision look right
4. Name 2 leverage points that would break the reinforcing loops, and say what
makes them leverage rather than just more effort
```

Prompt 6 — Divergent Ideation with GenAI (Breadth Before Judgement)

Use for: A5 · K4 — Generate a wide solution set using appropriate problem-solving techniques.

```
Act as an innovation consultant running a divergent ideation session. Do NOT
evaluate or filter yet.

Root cause to solve: A Singapore health-screening operator has 44% annual
turnover among permanent phlebotomists. Vacancies are backfilled by agency
relief staff who take 14 min per blood draw versus 6 min for trained staff. 40%
of the 07:00 roster is relief staff, causing 31% of morning appointments to
overrun and 23% of samples to miss the fixed 09:45 lab courier.

Constraints: S$250k over 12 months; MOH phlebotomy competency standards cannot
be relaxed; the 07:00-09:30 fasting window and the 09:45 courier are both
fixed.

Generate 20 distinct solutions grouped under:
- People & Retention
- Training & Onboarding
- Process & Scheduling
- Technology & Automation
- Customer/Client Experience

Rules:
- Include at least 4 ideas that would normally be dismissed as unrealistic
- Include at least 2 ideas borrowed from a COMPLETELY different industry (say
which)
- One line each, phrased as an action
- Do NOT rank, cost or evaluate them

Then, separately: apply SCAMPER (Substitute, Combine, Adapt, Modify, Put to
another use, Eliminate, Reverse) to the single assumption that 'every draw must
be done by a phlebotomist at the centre between 07:00 and 09:30' and give one
idea per SCAMPER letter.
```

Prompt 7 — Converging: Impact-Ease Matrix and Weighted Decision Matrix

Use for: A5 · K4, K6 — Shortlist and evaluate the most viable ideas using decision-making techniques.

```
Act as a decision analyst. I will give you a list of candidate solutions. Do
NOT add new ideas.

STEP 1 — Classify every solution on an Impact vs Ease matrix as exactly one of:
- Quick Win (high impact, easy)
- Big Bet (high impact, hard)
- Fill-In (low impact, easy)
- Thankless Task (low impact, hard)
Give a one-line reason for each placement.
```

```
STEP 2 — Take only the Quick Wins and Big Bets and score them in a weighted
decision matrix:
- Impact on root cause (weight 35%)
- Speed to measurable effect (20%)
- Cost within a S$250k envelope (20%)
- Clinical/regulatory risk (15%)
- Staff and client acceptance (10%)
Score each 1-5, show the weighted total, and rank them.

STEP 3 — Recommend the best FOUR as a portfolio, and explicitly state:
- why this combination is stronger than the top four individual scores
- which two solutions overlap or make each other redundant
- what the portfolio still leaves unaddressed

Solutions: [paste your team's list here]
```

Prompt 8 — Building the Corrective Action Plan

Use for: A4 · K9 — Develop corrective action plans for shortfalls identified.

```
You are an execution-focused planning assistant. Convert each approved solution
into a corrective action plan row. Be concrete and brief — no theory.

For EACH solution, produce exactly these fields:
- Root cause addressed (link back to the diagnosis)
- Corrective action (one specific sentence)
- Owner (a job title, not a department)
- Timeline (start, key milestone, completion)
- Resources (budget in S$, people, tools)
- Success measure (metric + baseline + target)
- Key risk + mitigation
- Review checkpoint (date and the decision to be made at it)

Then flag, in a separate section:
1. Any two actions that depend on each other and must be sequenced
2. Any owner who appears on more than two actions (overload risk)
3. Any success measure that cannot actually be measured with data the
organisation already has

Approved solutions: [paste your four here]
Root cause: 44% phlebotomist turnover leaving 40% of the 07:00 roster
untrained, causing 14-min draws vs 6-min, 31% appointment overruns and 23% of
samples missing the 09:45 courier.
Budget: S$250,000 over 12 months.
```

Prompt 9 — Implementation Planning and Stakeholder Resistance

Use for: A6 · K3, K8 — Draw up implementation plans and address factors affecting effectiveness.

```
Act as a change management and implementation planning advisor for a Singapore
fashion retailer.
```

```
Approved solutions: (1) one-page guest checkout; (2) CDN + caching to bring app
load from 4.8s to under 2.0s; (3) supplier QC checklist to cut the 12% return
rate; (4) automated cart-recovery notifications.

PART A — Implementation plan. For each solution give ONLY:
- Action (what will be done)
- Method/tool (how)
- Owner and timeline (who, by when)
- Sequencing note (what must happen before it)

PART B — Stakeholder strategy. Stakeholders and positions:
IT Development (at capacity, resists workload); Head of Marketing (sponsored a
S$180k campaign the diagnosis implies was misdirected — resists); Store
Operations (indifferent); 12 suppliers (face new QC standards — resist);
Customer Service (supportive, overloaded); Finance (conditional on ROI
evidence); COO (champion); frontline retail staff (fear job loss).

For EACH stakeholder give:
- Their specific objection in THEIR words, not yours
- What they need to hear or receive to move
- One concrete action to secure their cooperation
- Who is best placed to deliver that message and why

PART C — Name the single stakeholder most likely to sink this rollout, and
explain what makes them dangerous. Then give a 90-day communication plan: what
is communicated, to whom, how often, by whom.

Be blunt about political realities. Do not assume goodwill.
```

Prompt 10 — Measuring Effectiveness: Did It Actually Work?

Use for: A7 · K7 — Evaluate the effectiveness of implemented solutions and implementation plans.

```
Act as a performance evaluation analyst. Assess whether this improvement
programme worked. Be rigorous and do not flatter the results.

Results after 90 days (metric: baseline -> target -> actual):
App load 4.8s -> <2.0s -> 1.7s
Checkout completion 51% -> 68% -> 63%
Cart abandonment 74% -> 60% -> 64%
Online sales S$1.9m -> S$2.8m -> S$2.5m
CSAT 70% -> 85% -> 72%
Complaints/week 410 -> 150 -> 295
Product return rate 12% -> 5% -> 13% (WORSE)
Repeat purchase 29% -> 40% -> 31%
Supplier QC checklist adopted by 4 of 12 suppliers
Cart recovery conversion 3.1% against an 8% target

Context: the 90-day window included the Great Singapore Sale.

Produce:
```

```
1. A verdict per metric: Met / Partially met / Not met / Deteriorated
2. Which solutions can be credited with which movements — and where you CANNOT
establish attribution, say so explicitly and explain why
3. Why the product return rate got WORSE despite the QC initiative — give the
most likely explanation and state what evidence would confirm it
4. Separate LEADING from LAGGING indicators here, and explain why CSAT and
repeat purchase barely moved even though operational metrics improved
5. How the Great Singapore Sale confounds this evaluation, and what analysis
would isolate the programme effect from the seasonal effect
6. A clear recommendation to the CFO on releasing the next S$200k tranche:
continue, adjust, or stop — with reasoning
7. The three things you would measure differently next time
```

Preparing for the Assessment

Format

- Written Assessment (WA) — Short-Answer Questions (SAQ), open book.
- Case Study (CS) — an integrated workplace problem-solving scenario, open book.
- Open book covers the slides, Learner Guide and approved materials only.
- A minimum of 75% attendance based on the SSG Digital Attendance record, and a 'Competent' result in both instruments, are required for certification and funding.

What you should be able to do

- Write a six-element problem statement from a vague complaint plus evidence, and identify which metric represents the real performance deficiency.
- Run a 5 Whys chain to a systemic root cause and justify each level with evidence.
- Populate a fishbone across six dimensions and identify which dimension an intervention targets.
- Compute a Pareto cumulative percentage, identify the vital few, and state what Pareto cannot tell you.
- Identify reinforcing and balancing loops and explain why a fix rebounded.
- Generate a wide solution set and distinguish a hard constraint from a habit.
- Apply an Impact-Ease screen and a weighted decision matrix, including a sensitivity test.
- Write a corrective action plan with all eight components and a named owner per action.
- Predict stakeholder resistance and specify the move and the messenger for each.
- Evaluate results against baseline and target, state attribution honestly, and distinguish solution failure from implementation failure.

Common mistakes to avoid

- Writing a problem statement with no baseline number — this makes evaluation impossible later.

- Stopping a 5 Whys chain at a person ('the relief staff are slow') rather than a system condition.
- Leaving a fishbone bone empty and treating that as 'no causes' rather than 'no visibility'.
- Claiming a Pareto analysis proves an SLA is achievable without doing the arithmetic.
- Proposing 'hire more people' as a leverage point — that is effort, not leverage.
- Ranking solutions before the criteria and weights have been agreed and written down.
- Naming a department rather than a role as the owner of a corrective action.
- Assuming stakeholders will cooperate because the analysis is correct.
- Crediting a programme with every improvement while blaming the season for every shortfall.

Glossary

- **Baseline —** The measured value of a metric before any intervention — the reference point that makes evaluation possible.
- **Balancing loop (B) —** A feedback loop that pushes a system toward a target and stabilises it.
- **Cognitive fixedness —** The tendency to default to familiar solutions even when the context has changed.
- **Confounding —** Another factor changing during your measurement window, making attribution uncertain.
- **Consensus fatigue —** Accepting a weak option merely to end a long discussion.
- **Corrective action plan (CAP) —** A structured plan converting a chosen solution into owned, timed, measurable actions.
- **DMAIC —** Define, Measure, Analyze, Improve, Control — the Six Sigma improvement framework.
- **Fishbone (Ishikawa) diagram —** A cause-categorisation diagram organising potential causes across dimensions.
- **Five Whys —** Asking 'why' repeatedly, typically five times, to move from symptom to root cause.
- **Generic Parts Technique —** Stripping a problem to abstract functional components to break fixedness.
- **Hallucination —** A confident, plausible but unsupported output from a generative AI model.
- **IDEAL —** Identify, Define, Explore, Act, Look back — a five-stage problem-solving heuristic.
- **Ill-defined problem —** A problem with an unclear goal and no standard method — most workplace problems.
- **Impact-Ease matrix —** A 2x2 screen classifying options as Quick Wins, Big Bets, Fill-Ins or Thankless Tasks.

- **Lagging indicator —** A measure that reflects accumulated outcomes and responds slowly (e.g. CSAT).
- **Leading indicator —** A measure that responds quickly and signals whether a change is taking hold.
- **Leverage point —** A place where a small structural change produces a large, self-sustaining effect.
- **Pareto analysis —** Ranking causes by contribution to find the vital few responsible for most of the effect.
- **PDCA —** Plan-Do-Check-Act — an iterative improvement and control cycle.
- **Performance deficiency —** A measurable gap between required and actual performance in a system or process.
- **Problem statement —** A structured statement carrying context, metric, baseline, target, timeframe and constraints.
- **Prompt engineering —** Structuring instructions to a generative AI model to obtain reliable, usable output.
- **RAG —** Retrieval-augmented generation — grounding an AI model in your own data to reduce hallucination.
- **Reinforcing loop (R) —** A feedback loop that compounds, making things progressively better or worse.
- **Root cause —** The underlying system condition which, if removed, stops the problem recurring.
- **SCAMPER —** Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse.
- **Sensitivity test —** Re-ranking options under changed criteria weights to test how robust a decision is.
- **Symptom —** The visible effect of a problem, frequently mistaken for the problem itself.
- **Systems thinking —** Analysing a problem as a web of interacting feedback loops rather than a linear chain.
- **Weighted decision matrix —** A scoring tool applying agreed weights to agreed criteria to rank options defensibly.
- **Well-defined problem —** A problem with a clear goal and a known method of solution.